

ELECTRONIC IRRADIATION OF FOODS

An Introduction to the Technology

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CHAPTER 6

PRINCIPLES OF ELECTRON ACCELERATION IN MICROWAVE ACCELERATORS

In this chapter we discuss the essential concepts involved in the generation and acceleration of electron beams in microwave accelerator structures. Specifically, we discuss the motion of electrons in dc and oscillating fields, the formation of a beam in an electron gun, the properties of resonant electromagnetic cavities and coupled-cavity linac structures, and methods for coupling power from a microwave source into an accelerating structure.* Finally, with the aid of numerical simulations we illustrate explicitly how the confined microwave fields are used to accelerate electrons.

6.1. ELECTRON ACCELERATION

The time rate of change of the momentum $\mathbf{p}$ of a particle subject to a force $\mathbf{F}$ is given by Newton's second law according to $d\mathbf{p}/dt = \mathbf{F}$. For a particle of charge q , the force is given by the Lorentz law as

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

in which $\mathbf{E}$ and $\mathbf{B}$ denote the electric and magnetic field vectors, and $\mathbf{v}$ is the particle velocity vector. For an electron, $q = -e$, with the unit electron charge equal to 1.6×10^{-19} coulombs. Electrons have a small rest mass m of 9.1×10^{-31} kg. Consequently, they become relativistic with only modest acceleration, and it is necessary to use the relativistic particle momentum in describing their motion, with $\mathbf{p}$ related to the electron velocity by $\mathbf{p} = \gamma m \mathbf{v}$. $\gamma = (1 - \beta^2)^{-1/2}$ and $\beta = |\mathbf{v}|/c$ are the usual relativistic factors. The relativistic equation of motion for an electron is therefore

$$d(\gamma m \mathbf{v})/dt = -e(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \quad (6.1)$$

It is instructive to take the vector dot product of the electron velocity with both sides of Eq. (6.1). The result is that

$$d(\gamma mc^2)/dt = -e \mathbf{v} \cdot \mathbf{E} \quad (6.2)$$

From the special theory of relativity the quantity in parentheses is identified as the total energy of the electron. Since the rest mass energy of the electron is $mc^2 = 0.511$ MeV, the energy associated

* Only normally conducting structures will be considered. At the present time, the high capital cost of superconducting technology is not justified for the low electron kinetic energies appropriate for food irradiation.

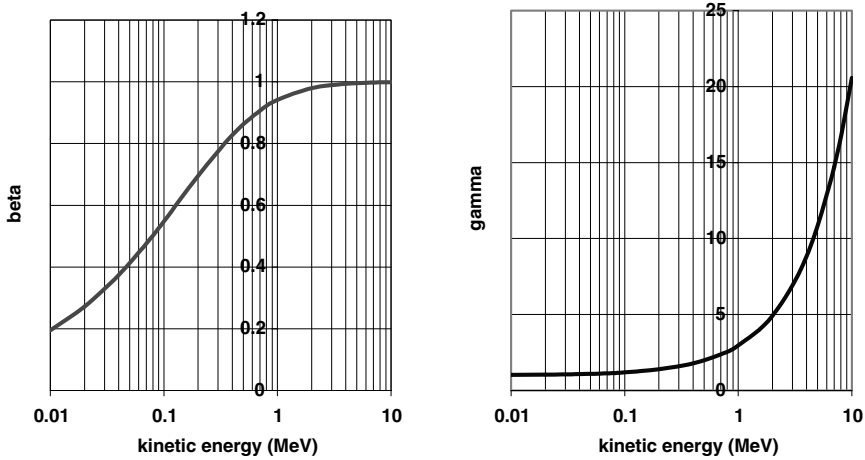


Figure 6-1. Variation of the relativistic beta and gamma factors with electron kinetic energy.

with the electron motion, the kinetic energy, must be $(\gamma - 1)mc^2$. The variation of γ and β with electron kinetic energy are shown in Figure 6-1. Note also that only an electric field in the direction of the electron motion can change its kinetic energy. Since the vector formed by $\mathbf{v} \times \mathbf{B}$ is necessarily perpendicular to the velocity, a static magnetic field can only deflect an electron.

6.1.1. Acceleration in a Constant, Uniform Electric Field

We first consider the acceleration of an electron in an acceleration gap of width d to which a constant voltage V_0 has been applied. The geometry is schematically shown in Figure 6.2. The electron is assumed to enter the gap at time $t = 0$ with an initial velocity v_0 (and kinetic energy $(\gamma_0 - 1)mc^2$) in the direction opposite the electric field vector of magnitude $E_0 = V_0/d$. Eq. (6.1) therefore becomes

$$d(\beta\gamma)/dt = (e/mc)E_0$$

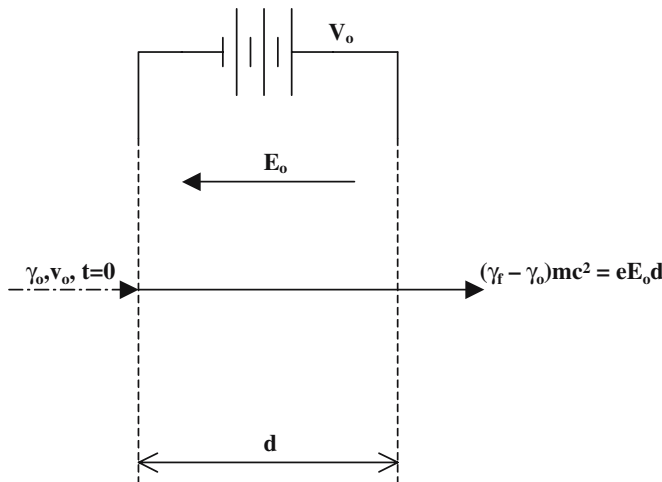


Figure 6-2. Electron acceleration in a constant, uniform electric field.

Performing the indicated integration yields

$$\beta\gamma = \beta_0\gamma_0 + (e/mc)E_0t \quad (6.3)$$

When relativistic effects are important, it is the quantity $\beta\gamma = (\gamma^2 - 1)^{1/2}$ that increases linearly with time, not the electron velocity.

Recognizing that the kinetic energy gain on crossing the gap is eE_0d , Eq. (6.3) can be used to determine the electron transit time. If the electron enters the gap with an initial kinetic energy in excess of 1 MeV, the transit time τ is essentially d/c . At the other extreme, if the initial velocity is nearly zero, then the transit time is approximately

$$\tau = (d/c)[1 + 2mc^2/(eE_0d)]^{1/2} \quad (6.4)$$

6.1.2. Acceleration in Oscillating Fields

We now consider the case in which the applied voltage is oscillatory, and the electric field amplitude can be represented as

$$E_z = E_0 \sin \omega t \quad (6.5)$$

Assuming that the electron velocity does not change appreciably while crossing the gap, then the kinetic energy gain (or loss) from Eq. (6.2) depends on the entry and exit times t_0 and t_1 according to

$$\Delta\gamma mc^2 = (evE_0/\omega)[\cos \omega t_1 - \cos \omega t_0] \quad (6.6)$$

Referencing t_0 and t_1 to the time t_c that the electron is at the center of the cavity, we have $t_0 = t_c - d/(2v)$ and $t_1 = t_c + d/(2v)$, and the term in square brackets can be written as

$$\cos(\omega t_c + \theta_g/2) - \cos(\omega t_c - \theta_g/2)$$

with the gap transit angle defined as $\theta_g = \omega d/v$. Using the trigonometric identity $\cos(a + b) = \cos(a)\cos(b) - \sin(a)\sin(b)$, Eq. (6.6) can be written as

$$\Delta\gamma mc^2 = eE_0dT \sin(\omega t_c) \quad (6.7)$$

with T , the transit time factor of the gap, defined as

$$T = [\sin(\theta_g/2)]/(\theta_g/2) \quad (6.8)$$

Maximum acceleration occurs when $\omega t_c = \pi/2$, which can nearly always be arranged for electrons; however, because of the transit time factor, the energy gain in the sinusoidally varying electric field is less than the energy gain (eE_0d) in the corresponding dc field. The transit time factor is plotted versus (θ_g/π) in Figure 6-3. As a numerical example, for a 3-GHz pillbox cavity having a width of 5 cm, the transit angle for a relativistic electron is equal to π , and the transit time factor has a value of only 0.64.

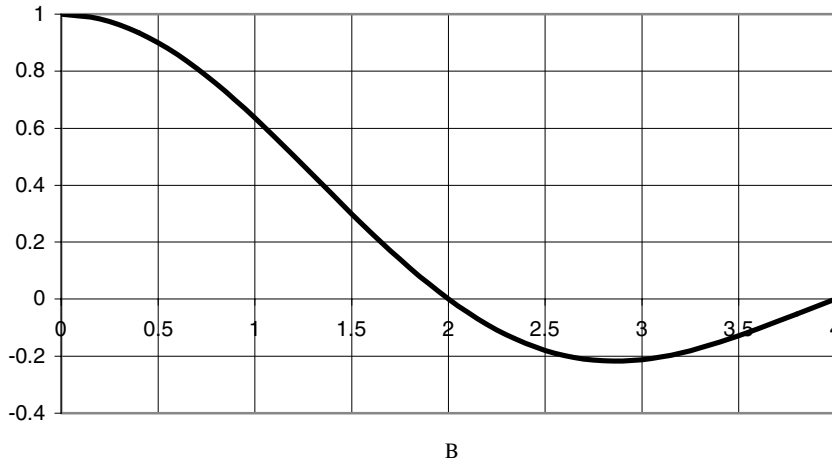


Figure 6-3. Transit time factor as a function of normalized transit angle.

6.2. ELECTRON BEAM GENERATION¹

The beam current in a microwave accelerator is relatively modest (typically of the order of one ampere or less), and is therefore almost always supplied by a thermionic cathode in a cylindrically symmetric electron gun configuration, as schematically shown in Figure 6-4. Electrons emitted from the cathode surface are accelerated by an applied voltage of typically several tens of kilovolts and geometrically focused through an aperture in a grounded anode. If it is necessary to control the beam current without varying the anode-cathode (A-K) voltage, a control grid may be interposed between the cathode and anode.

The factors that are primarily responsible for determining the cathode emission characteristics include the energy band structure of the material, the temperature, the magnitude of an applied electric field, and the influence of the electron space charge in the diode region. The details of the resulting electron flow depend on the diode geometry, the applied voltage and the presence or absence of a

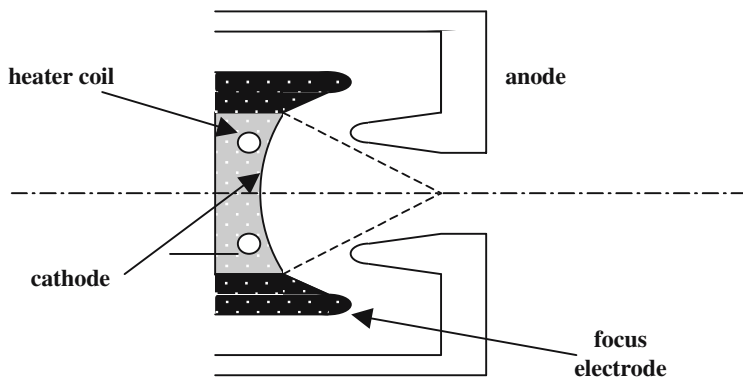


Figure 6-4. A cross-sectional view of a Pierce-type convergent diode electron gun² consisting of a spherically shaped cathode with an embedded heater coil, a focus electrode, and a specially-shaped anode with a central aperture. The angle between the outermost electron trajectory and the focus electrode is 67°.

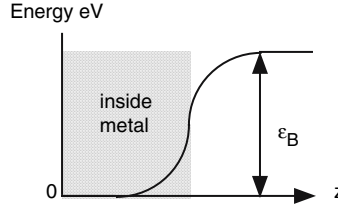


Figure 6-5. The metal interior can be conveniently regarded as an equipotential volume, with the zero energy level corresponding to the bottom of the conduction band. The potential barrier of height ϵ_B prevents electron emission from the surface. The magnitude of ϵ_B equals the sum of the Fermi level plus the work function.

magnetic field. In this section we briefly summarize the important features of the emission process, operation in temperature-limited and space-charge limited regimes, and the resulting electron flow.

6.2.1. Thermionic Emission

It is convenient to consider the interior of a metal as an equipotential volume with a potential barrier at the surface, as suggested in Figure 6-5. Since only differences in potential have physical meaning, we can assume that the zero energy level represents the bottom of the conduction band. The kinetic energies of the free electrons in the conduction band are therefore always positive with respect to this level.

Since electrons obey the Pauli exclusion principle, the probability that an electron has a particular energy ϵ is specified by the Fermi-Dirac distribution function given by

$$f(\epsilon) = \{1 + \exp[(\epsilon - \epsilon_f)/kT]\}^{-1} \quad (6.9)$$

in which $k = 1.38 \times 10^{-23}$ J/°K is the Boltzmann constant, T is the absolute temperature (°K), and ϵ_f is termed the Fermi level.

The energy difference between the Fermi level and the barrier height is called the work function ϵ_w ; ϵ_w lies in the range of 2–6 eV for most metals. Emission of an electron from the surface can occur if the electron kinetic energy exceeds $\epsilon_B = \epsilon_f + \epsilon_w$. At temperatures above absolute zero there are always some electrons having energies greater than ϵ_f . With sufficient heating it is therefore possible to create a significant electron population with energies in excess of $\epsilon_w + \epsilon_f$.

The current density j produced by heating the metal can be computed by integrating the electron flux incident on the metal surface from the interior over all energies exceeding the surface potential barrier. The result is

$$\begin{aligned} j &= -(4\pi emk^2/h^3)T^2 \exp(-\epsilon_w/kT) \\ &= -1.2 \times 10^6 T^2 \exp(-\epsilon_w/kT) \text{ amps/m}^2 \end{aligned} \quad (6.10)$$

This equation is known as the Richardson-Dushman equation for thermionic emission. In practice it is found that the temperature dependence of thermionic emission closely follows Eq. (6.10), but that the value of the numerical constant is typically a factor of two to four lower. This decrease is usually attributed to two factors: (1) a variation of the work function over the emission surface, and (2) a variation in the work function with temperature.

Eq. (6.10) indicates that obtaining high electron emission rates requires high temperatures and low work functions. Unfortunately, the pure metals that have low work functions are unusable because

they also have low melting temperatures. It is possible, however, to reduce the work functions of some high-melting-temperature metals, such as tungsten and osmium, by introducing surface impurities such as barium and oxygen. While complex, the surface chemistry responsible for reducing the work function is now reasonably well understood. The oxides most often used are BaO, SrO, and CaO. To make these coating materials suitable for use they must first be “activated”, i.e., heated to a temperature somewhat higher than the normal operating temperature. During this process, the oxides are partially reduced giving rise to free barium.

Such metal oxide cathodes have proven to be quite useful, delivering current densities on the order of 1 amp/cm² at relatively low temperatures (800 °C), but they do have drawbacks: (1) the coating is usually quite resistive, (2) they tend to blister, (3) they can be destroyed by back-bombardment, and (4) they can be easily poisoned by residual gases. These limitations have led to the development of the so-called dispenser cathode. Although there are many varieties of dispenser cathode, the common feature is a porous tungsten matrix through which barium can diffuse to the surface. For example, in the Philips B-type cathode the porous tungsten is impregnated with a compound of BaO, CaO and Al₂O₃ in the ratio of 5:3:2. With a mix of 4:1:1, the cathode is known as an S-type cathode. These cathodes are rugged, and can provide an emission density of several tens of amps/cm² when operated at temperatures of about 1100 °C. The lifetime of a dispenser cathode depends on the availability of free barium that can migrate to the surface, which in turn depends on the cathode temperature. A thin surface overcoating of osmium, iridium, or ruthenium can lower the work function by almost 20%. This reduces the operating temperature for a given current density by almost 100 °C, resulting in extending the cathode life by about a factor of ten.

6.2.2. Space-Charge-Limited Operation

The characteristic variation of the current density from a thermionic cathode is plotted as a function of temperature in Figure 6-6. At low temperatures the current increases with temperature according to the Richardson-Dushman law and is said to be temperature-limited (TL). As the temperature is further increased, however, the current becomes nearly independent of temperature. The phenomenon that limits the current is the electron space charge in the diode. Electron guns in microwave accelerators are almost always operated in this space-charge-limited (SCL) mode because it eliminates the need for precise temperature and work function uniformity over the cathode surface. It also eases voltage and current stability requirements for the cathode heater.

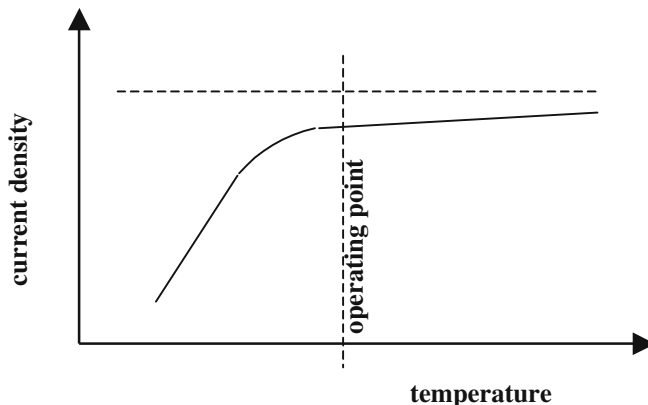


Figure 6-6. Variation of the emission current density from a thermionic cathode as a function of temperature. The operating point is usually chosen to be in the flat, space-charge-limited region of the curve.

The space-charge-limiting current for a particular electron gun geometry can be derived by assuming that the electric field *vanishes* at the cathode surface, and that the emitted electrons have zero initial velocity. The velocity of an electron at any position between the electrodes can then be determined from energy conservation. Solving Poisson's equation for a one-dimensional, planar, non-relativistic electron gun, the resulting current density is given as

$$\begin{aligned} j &= -(4/9)\epsilon_0(2e/m)^{1/2}(V^{3/2}/d^2) \\ &= -2.33 \times 10^{-6}(V^{3/2}/d^2)(\text{amps}/\text{m}^2) \end{aligned} \quad (6.11)$$

Eq. (6.11) is the familiar one-d Child-Langmuir law; it is accurate to within 10% for voltages up to approximately 1 MeV.

6.2.3. Electron Gun Design

Multiplying Eq. (6.11) by the effective cathode emission area A , the Child-Langmuir equation is commonly written in the form

$$I = PV^{3/2} \quad (6.12)$$

with $I = jA$. P , the diode perveance, is a numerical factor that depends only on the diode geometry. For the low current guns used in microwave accelerators, the gun is usually based on a convergent design due to Pierce, in which the electron flow is assumed to occur between an outer sphere (the cathode) and an inner sphere (the anode nose cone). Let the radius of curvature of the cathode be R_c , and the equivalent radius of the anode nose cone be R_a . Further, let θ denote the half-angle arc of the spherical cathode. Then the perveance is approximately given by

$$P = 14.7 \times 10^{-6}(1 - \cos \theta)\alpha^{-2} \quad (6.13)$$

where $\alpha = a + 0.3a^2 + 0.075a^3$, and $a = \ln(R_c/R_a)$. As a numerical example, if $\theta = 12$ degrees and $R_a = 0.4R_c$, then $P = 2.1 \times 10^{-7}$. An applied voltage of 25 kV would therefore yield a beam current of about 0.84 amps.

A numerical solution using the EGUN code³ for the electron trajectories in an electron gun with geometrical parameters similar (but not identical) to those above is shown in Figure 6-7. The beam current is 1.33 amps at an applied voltage of 32.5 kV, giving a perveance of 2.3×10^{-7} .

6.3. RESONANT ELECTROMAGNETIC CAVITIES

Electromagnetic radiation is characterized by frequency f and wavelength λ , with the two parameters being related by the speed of light c according to $f = c/\lambda$. With $c = 3 \times 10^8$ m/s, the wavelength in centimeters is given by $\lambda = 30/f$, with f in gigahertz (GHz). The several regions of the electromagnetic spectrum are described in Figure 6-8, according to frequency and wavelength, and origin.

The longest wavelengths (radio waves) are produced by oscillating electric currents, while infrared wavelengths (heat) are produced by molecular vibrations. The term "microwave" generally describes that portion of the short radio waves having frequencies ranging from about 300 MHz to about 100 GHz. The narrow visible and ultra-violet wavelengths are produced by a rearrangement of the electrons in outer atomic shells. X-rays can be produced by rearrangements of electrons in

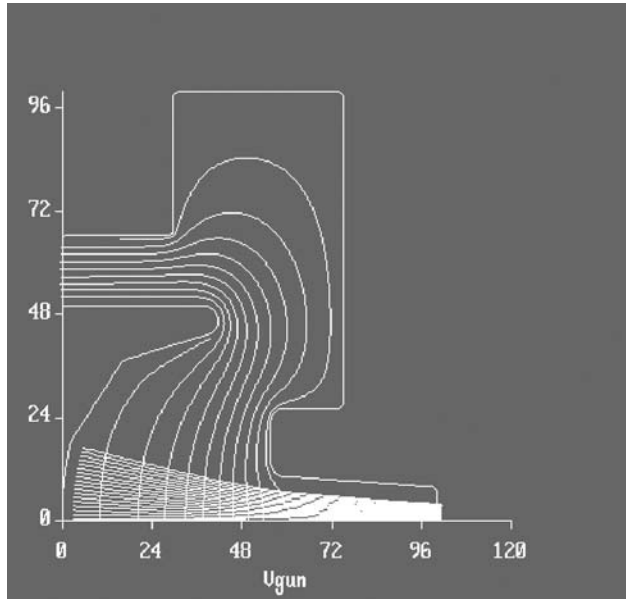


Figure 6-7. Electron flow trajectories in a space-charge-limited electron gun, as computed by the EGUN code. With 32.5 kV applied voltage, the gun generates 1.33 amps.

the inner atomic shells and by the abrupt slowing of energetic electrons (bremsstrahlung, or braking radiation). Gamma rays are produced by the de-excitation of radioactive nuclei.

Electromagnetic waves in the radio-frequency range can be confined in metallic cavities, and this fact permits the construction of radio-frequency (rf) accelerators. This is achieved using resonant electromagnetic cavities and coupled-cavity linac structures. In the remainder of this section we consider resonant cavities; in the following section we will consider coupled-cavity linac structures.

6.3.1. Maxwell's Equations

The underlying principle of electromagnetic radiation is that a time-changing magnetic field will induce an electric field, and a time-changing electric field will induce a magnetic field. The mathematical expression of this symmetry is embodied in the time-dependent Maxwell equations. Assuming vacuum conditions and no free charges or currents, the differential form of these equations is given below, with bold-faced type denoting vector quantities:⁴

$$\nabla \times \mathbf{E} + \partial \mathbf{B} / \partial t = 0 \quad (6.14)$$

$$\nabla \times \mathbf{H} - \partial \mathbf{D} / \partial t = 0 \quad (6.15)$$

The variables in these equations are identified as follows:

- E** is the electric field intensity in volts/meter,
- D** is the electric displacement in coulomb/meter²,
- B** is the magnetic induction in weber/meter², and
- H** is the magnetic field intensity in amperes/meter.

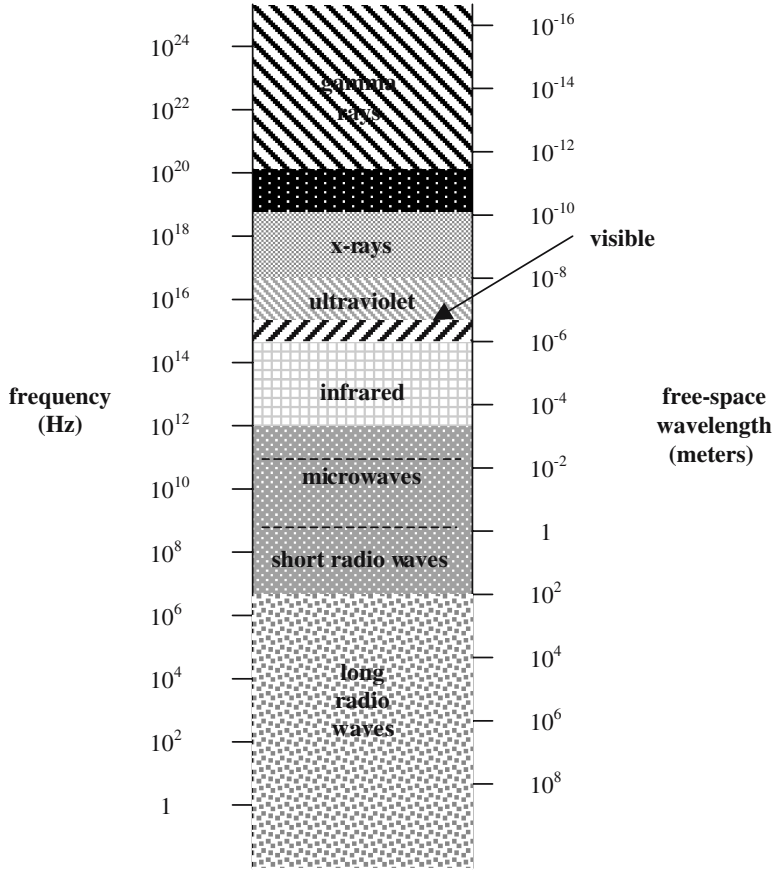


Figure 6-8. The electromagnetic spectrum.

In a vacuum, these fields are related according to $\mathbf{D} = \epsilon_0 \mathbf{E}$, and $\mathbf{B} = \mu_0 \mathbf{H}$, with $\epsilon_0 = 8.85 \times 10^{-12}$ farads/meter being the permittivity of free space, and $\mu_0 = 1.26 \times 10^{-6}$ henry/meter being the permeability of free space. It may be verified that the speed of light $c = (\epsilon_0 \mu_0)^{-1/2} = 3 \times 10^8$ m/s.

Assuming a time dependence of the fields in the form of $e^{j\omega t}$, the time derivative can be replaced by $j\omega$. In the frequency domain, Maxwell's equations then become

$$\nabla \times \mathbf{E} = -j\omega\mu_0 \mathbf{H}$$

$$\nabla \times \mathbf{H} = j\omega\epsilon_0 \mathbf{E}$$

Taking the curl of Eq. (6.14) on both sides, and using Eq. (6.15) for the curl of $\mathbf{H}$ gives

$$\nabla \times \nabla \mathbf{E} = -j\omega\mu_0 \nabla \times \mathbf{H} = \omega^2(\epsilon_0 \mu_0) \mathbf{E} \quad (6.16)$$

The vector identity for the curl of the curl of the electric field vector is expressed by

$$\nabla \times \nabla \times \mathbf{E} = -\nabla^2 \mathbf{E} + \nabla(\nabla \cdot \mathbf{E}) \quad (6.17)$$

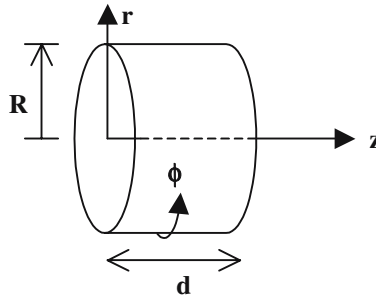


Figure 6-9. Coordinate system for a cylindrical cavity resonator.

In free space there is no free charge density and the latter term is zero. Therefore, the wave equation for the electric field becomes

$$\nabla^2 \mathbf{E} = -\omega^2(\epsilon_0 \mu_0) \mathbf{E} = -(\omega/c)^2 \mathbf{E} \quad (6.18)$$

A similar equation can be derived for the magnetic field intensity.

6.3.2. Normal Modes of a Pillbox Cavity⁵

These electromagnetic waves can be confined in high-conductivity cavities, with wave energy being continuously interchanged between the magnetic and electric fields. There are normal modes of oscillation that depend on the boundary conditions, and each mode has a definite resonant frequency. Consider the cylindrical (pillbox) cavity shown in Figure 6-9, having radius R and width d . In cylindrical coordinates, the “del squared” operator of Eq. (6.18) becomes

$$\nabla^2 = r^{-1} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \right) + r^{-2} \frac{\partial^2}{\partial \phi^2} + \frac{\partial^2}{\partial z^2} \quad (6.19)$$

Of particular interest for particle acceleration are those normal modes having an electric field that is a maximum along the axis of symmetry. Solutions to Eq. (6.18) for the axial component E_z that satisfy the boundary condition $E_z = 0$ at $r = R$ are the Bessel functions J_n of the first kind of order n . Sinusoidal functions satisfy the differential operator in the axial and azimuthal coordinates. It may be verified that the functions given in Eq. (6.20) are solutions of Eq. (6.18). It may also be verified that for these solutions the axial component of the magnetic field is zero; these modes are therefore designated as transverse magnetic, or TM_{npq} modes. There is a complementary set of TE modes.

$$E_z = E_{oz} J_n(x_{np}r/R) \cos(n\phi) \cos(q\pi z/d) \quad (6.20)$$

where

- $n = 0, 1, 2, \dots$ is the number of the periodicity in the azimuthal direction
- $p = 1, 2, 3, \dots$ is the number of zeros of the field in the radial direction
- $q = 0, 1, 2, \dots$ is the number of zeros of the field in the axial direction
- E_{oz} = amplitude of the electric field

Table 6.1. p^{th} Zeros of $J_n(x_{np}r/R)$ for TM_{np} Modes

p	n = 0	1	2
1	2.405	3.832	5.136
2	5.520	7.106	8.417
3	8.645	10.173	11.620
4	11.792	13.324	14.796

The resonant frequencies of the TM normal modes are determined by substituting Eq. (6.20) into Eq. (6.18). The result is (recall that $\omega = 2\pi f$)

$$\begin{aligned} (x_{np}/R)^2 + (q\pi/d)^2 &= (\omega/c)^2 \quad \text{or,} \\ f &= (c/2\pi)[(x_{np}/R)^2 + (q\pi/d)^2]^{1/2} \end{aligned} \quad (6.21)$$

The lowest zeros of the $J_n(x_{np}r/R)$ Bessel functions are summarized in Table 6.1. Of special interest are those TM modes having no variation in either the azimuthal or axial directions ($n = q = 0$). The TM_{010} mode, for which E_z is zero only at $r = R$, has the lowest frequency of these modes; from Table 6.1, it is given by

$$f_{010}(\text{Hz}) = 2.405c/(2\pi R) = 1.15 \times 10^{10}/R(\text{cm}) \quad (6.22)$$

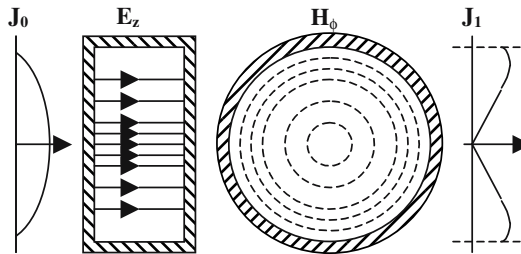
It is easily shown from Eq. (6.16) that the other field components of the TM_{010} mode are given as $E_r = E_\phi = H_r = H_z = 0$, and

$$H_\phi = jE_{oz}(\epsilon_0/\mu_0)^{1/2} J_1(x_{01}r/R) \quad (6.23)$$

These field distributions are shown schematically in Figure 6-10.

For a cavity oscillating in a particular normal mode the electric and magnetic energies are equal and in time quadrature, meaning that when the electric energy is a maximum, the magnetic energy is zero, and vice versa. The total electromagnetic energy stored in the cavity can be obtained by integrating either the electric or magnetic energy density over the cavity volume. For the TM_{010} mode of the pillbox cavity this integration gives

$$U = (\pi\epsilon_0/2)E_{oz}^2(dR^2)J_1^2(x_{01}r/R) \quad (6.24)$$

**Figure 6-10.** TM_{010} electric and magnetic field distributions in a cylindrical cavity.

Metals at room temperature have finite conductivity. A practical consequence is that the fields (and the surface currents that support them) can penetrate into the surface a characteristic distance, termed the “skin depth”, giving rise to resistive losses. This average power loss can be evaluated by integrating the magnetic power density over the inner surface of the conductor. For the TM_{010} mode of the pillbox cavity this yields

$$P_c = \pi R(d + R)(\epsilon_0/\mu_0)\rho_s E_{oz}^2 J_1^2(x_{01}r/R) \quad (6.25)$$

with ρ_s denoting the surface resistivity. A measure of the relative energy damping of a resonant cavity is the term “Q”, or quality factor, defined as follows:

$$Q = \omega \times (\text{stored energy/average power loss})$$

From Eqs. (6.24) and (6.25), the Q of the pillbox resonator is therefore given by

$$Q = (\mu_0\pi fR)/[\rho_s(1 + R/d)] = 1.2\mu_0c/[\rho_s(1 + R/d)] \quad (6.26)$$

The cavities of practically all microwave accelerators are made of high-conductivity copper which has a surface resistivity approximately given by $\rho_s = 2.61 \times 10^{-7}f^{1/2}$, with ρ_s in ohms for f in Hz.⁶ Therefore, the Q of a copper pillbox cavity resonating in the TM_{010} mode is approximately given as

$$Q = 1.73 \times 10^9 f^{-1/2}(1 + R/d)^{-1} \quad (6.27)$$

For practical dimensions, Q is of the order of 10,000. Q always increases with increasing d. Also, since the frequency varies inversely with radius, the quality factor increases with frequency for large R/d. However, for R/d comparable to unity or less, Q slowly decreases with frequency. As a numerical example, we assume a frequency of 3 GHz, with a corresponding cavity radius of 3.83 cm. For a cavity width of 5 cm,* which would correspond to a nominal voltage gain of 500 kV at an average field stress of 100 kV/cm, the resulting quality factor is approximately 17,900. While this value is reasonably high, it can be further increased by shaping the cavity surfaces, as discussed in Section 6.3.4.

6.3.3. Circuit Model of a Resonant Cavity⁵

The oscillating character of the energy stored in the electric and magnetic fields in the cavity is closely analogous to the oscillations in a resonant LC circuit in which energy is alternately stored in the magnetic field of an inductor and the electric field of a capacitor. In fact, for a cavity oscillating in a single mode, the simplest model for describing the behavior of a resonant cavity is provided by a parallel RLC circuit driven by a current source, as schematically shown in Figure 6-11. The driving current $I(t)$ produces a voltage $V(t)$ across the parallel combination of inductance L, capacitance C, and shunt resistance R, that can be identified with the axial voltage (E_0d) in the cavity.

Equating voltages and summing the currents leads to the second order circuit equation

$$dI/dt = V/L + (dV/dt)/R + C(d^2V/dt^2) \quad (6.28)$$

* 5 cm is also the distance a relativistic electron will travel in half the oscillation period.

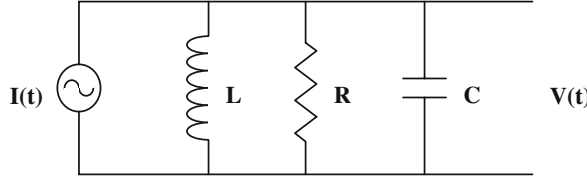


Figure 6-11. Parallel RLC model of a resonant cavity driven by a current source.

We note that the stored energy of the circuit is given by $U = CV_o^2/2$, while the dissipated power is $P = V_o^2/(2R)$, with V_o denoting the peak voltage.* Therefore, the Q of this circuit is just $Q = \omega_o RC$, with the resonant frequency defined as $\omega_o = (LC)^{-1/2}$.

Assuming a harmonic time dependence of the form $e^{j\omega t}$, Eq. (6.28) can be rewritten as

$$V = j\omega CI / [(\omega_o^2 - \omega^2) + j\omega\omega_o/Q]$$

and the shunt impedance of the circuit can be written as

$$Z = V/I = R(e^{j\phi})(1 + y^2)^{-1/2} \quad (6.29)$$

where $\phi = -\tan^{-1}y$, and the “detuning factor” y is defined by

$$y = Q(\omega/\omega_o - \omega_o/\omega) \quad (6.30)$$

The normalized shunt impedance Z/R is graphed in Figure 6-12, assuming $Q = 20,000$ and $\omega_o/2\pi = f_o = 3$ GHz.

The width of the resonance is defined as the difference between the frequencies at which the voltage falls to $2^{-1/2}$ of its maximum (the half-power points), which occurs when $y = 1$. Let $\delta\omega = |\omega - \omega_o|$ be the frequency difference corresponding to the half-power points. In the limit of large Q , y is approximately equal to $2Q(\delta\omega/\omega_o)$. Therefore, the resonance width $\Delta\omega$ can also be written in terms of the quality factor as

$$\Delta\omega = 2\delta\omega = \omega_o/Q \quad (6.31)$$

Consequently, measuring the resonant response of a microwave cavity provides a convenient method for determining the quality factor. For the example of Figure 6.12, the resonance width of the unloaded cavity is 150 kHz.

The general solution of Eq. (6.28) is the sum of a homogeneous solution and the particular solution, and can be written as⁷

$$V(t) = (ae^{j\omega_1 t} + be^{-j\omega_1 t})e^{-\omega_o t/(2Q)} + ZI(t) \quad (6.32)$$

where a and b are constants determined from the initial conditions, and

$$\omega_1 = \omega_o[1 - (2Q)^{-2}]^{-1/2}$$

* The accelerating cavity shunt impedance is usually defined as $R_s = 2R$.

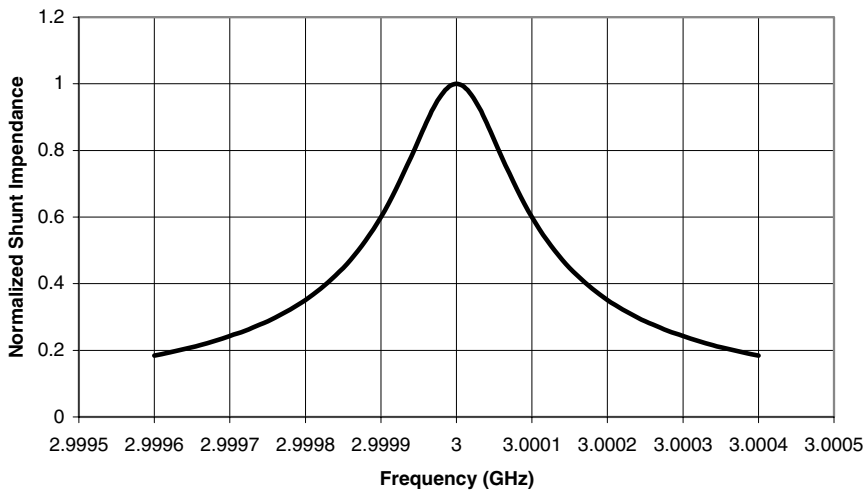


Figure 6-12. Response curve of a resonant cavity with $f_0 = 3$ GHz and $Q = 20,000$.

At resonance, $y = \phi = 0$, and in the limit of large Q , $\omega_1 = \omega_0$. If a driving current I_0 is applied at $t = 0$, when the voltage is initially zero, then $a = b = -RI_0/2$. Taking the real part of Eq. (6.32), the voltage is given by

$$V(t) = RI_0[1 - e^{-t/\tau}] \cos(\omega_0 t) \tag{6.33}$$

with the time constant of the resonator being defined by $\tau = 2Q/\omega_0$. The voltage is oscillatory, and the amplitude of the oscillation increases over a few time constants to the value RI_0 . If the driving current is abruptly reduced to zero, the voltage will also decrease to zero with the characteristic time constant τ .

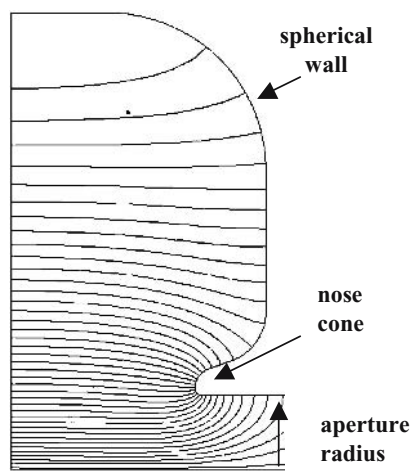


Figure 6-13. Generic shape of an accelerating cavity “half-cell” with the field lines as calculated by SUPERFISH.

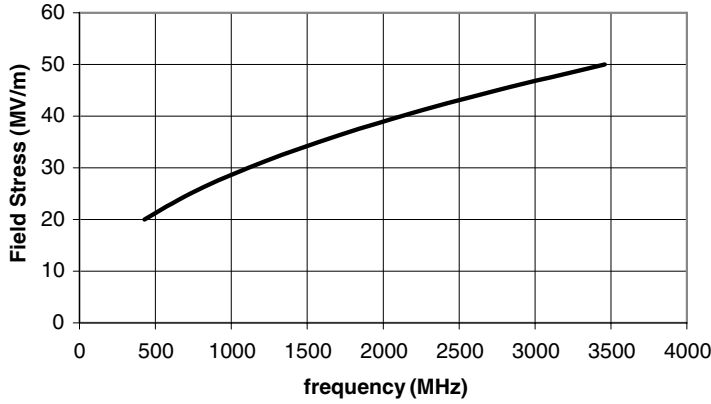


Figure 6-14. The limiting electric field stress, as given by the Kilpatrick criterion, Eq. 6.35.

6.3.4. Accelerating Cavity Design Issues

There are several factors to consider when designing resonant cavities for particle acceleration. In addition to resonating in the correct mode at the desired frequency, the transit time factor, shunt impedance, and peak field stress are all important design parameters. Recall that the shunt impedance is a measure of the effectiveness of producing an axial voltage $V = E_0 d$ for a given power dissipation. For particle acceleration it is usually more important to maximize the particle energy gain per unit of power dissipation. Since the shunt impedance is defined in terms of the cavity power loss and voltage as $R = V^2/P$, the effective shunt impedance can be obtained by replacing V by (VT) to yield

$$R_e = (VT)^2/P = RT^2 \quad (6.34)$$

The effective shunt impedance per unit length is similarly given as $Z_e = ZT^2$.

The general shape of an accelerating cavity is shown in Figure 6-13. Beginning with the simple pillbox geometry, nose cones are added to concentrate the fields in the gap region, thereby increasing the transit time factor. Also, the outer walls are given a spherical shape, corresponding to the smallest surface area for a given volume, in order to increase the quality factor and the shunt impedance. The aperture of the nose cone must be large enough to guarantee low beam loss, although the transit time factor decreases with increasing aperture.

To avoid electrical breakdown (sparking) on the nose cones, it is necessary to limit the peak electrical stress. The Kilpatrick criterion,⁸ given as Eq. 6.35 with the frequency f in MHz and E_k the limiting field stress in MV/m, is often used for this purpose.

$$f = 1.64E_k^2 e^{-8.5/E_k} \quad (6.35)$$

The Kilpatrick criterion is graphed in Figure 6-14. Note that E_k increases with frequency. The peak design stress level is usually expressed as $E_s = bE_k$, with b termed the “bravery factor.” Since the Kilpatrick criterion is based on older experimental data obtained under relatively poor vacuum conditions (by today’s standards), b is typically 1.5 or greater, especially for pulsed systems with

Table 6.2. Typical SUPERFISH Cavity Design Output Data

Parameter	Value
Resonant Frequency	1318.35 MHz
Transit Time Factor (T)	0.80
Quality Factor (Q)	25861
Shunt Impedance per Unit Length	84.45 M Ω /m
ZT ²	54.5 M Ω /m
Peak to Average Field Ratio	3.82

macropulses less than one millisecond. Thus, for operation at 1.3 GHz, the design field stress is typically 45 MV/m, increasing to perhaps 65 MV/m at 3 GHz.*

Values of the key cavity design parameters can be obtained by solving Maxwell's equations subject to the appropriate boundary conditions, and there are now several good computer programs that perform these calculations, e.g., SUPERFISH.⁹ As an example, the SUPERFISH output for a 1.3-GHz version of the cavity shown in Figure 6-13 is summarized in Table 6.2. Coupling slots (discussed in the next section) would lower the operating frequency by about 18 MHz. In practice, the quality factor as calculated by SUPERFISH is usually reduced by about 15% to account for surface irregularities, coupling slots, etc. In addition, a rule-of-thumb is that the shunt impedance will decrease by about 3% for every 1% increase in the coupling constant.

The nose cone shape used in Figure 6-13 produces a peak-to-average field ratio of 3.82. Assuming $b = 1.5$, and using a Kilpatrick limit of 32 MV/m from Figure 6-14, the maximum design field stress for this cavity is 48 MV/m, implying an average accelerating field of 12.5 MV/m. Multiplying by the transit time factor $T = 0.8$ from Table 6.2 gives a maximum effective gradient of 10 MV/m.

6.4. MULTIPLE-CAVITY ACCELERATING STRUCTURES

Multiple-cavity accelerators can be constructed from a series of independent cavities, with each driven by its own separate microwave source. Provided that the phase of the fields in each cavity can be independently adjusted so as to maintain phase synchronism of the fields with the accelerated bunches, such an approach allows for very flexible, and perhaps, high-average-power operation. With one accelerating gap per cavity, the cavity spacing in such an accelerator is simply $d = cT/2$ for relativistic electrons, with T being the microwave period. In fact, such independent-cavity linacs have been built, primarily for heavy ion acceleration (the UNILAC,¹⁰ for example).

6.4.1. Comparisons of TW and SW Structures

With the powerful, pulsed microwave sources now available, however, it is usually far simpler to construct linear accelerating sections consisting of many coupled cavities, with each section driven by a single microwave source. There are two such structures in common usage: the traveling-wave (TW) linac, and the standing-wave (SW) linac. The TW structure consists of a cylindrical waveguide that is periodically loaded with conducting disks, as schematically shown in Figure 6-15. The microwave

* A resonant breakdown phenomenon termed electron "multipactoring" (implying multiple electron impacts) is almost never an issue for the accelerating cavities in high-gradient electron microwave accelerators, but it can be a problem in the low energy sections of ion accelerators, and in the low-field coupling cavities of electron linacs. It has been noted that the Kilpatrick voltage limit is approximately 2000 times the electron multipactor condition, suggesting that proton multipactoring might be the limiting mechanism in high-voltage rf breakdowns. See Reference 18 for more details.

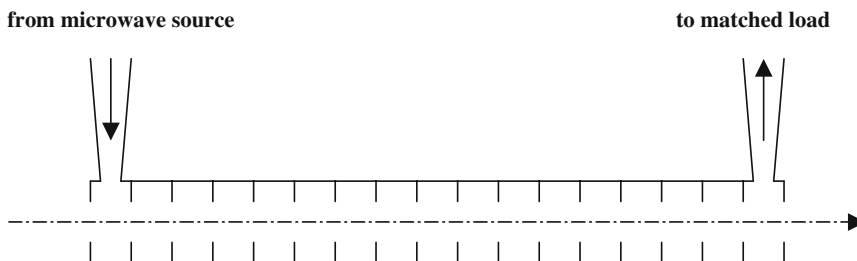


Figure 6-15. Schematic representation of a traveling wave linear accelerator.

power travels in one direction, from input to output. Internal reflections from the periodic disks reduce the wave phase velocity below the speed of light, enabling particle acceleration as the waves and particle bunches travel synchronously down the structure. The unused microwave power at the high-energy end is usually coupled into a matched resistive load. TW linacs can be designed to operate in either constant impedance or constant gradient modes. If the accelerating cavities defined by the disks are identical, then the impedance per unit length is constant, and the accelerating gradient decreases as the transmitted power droops. If a constant gradient is desired, the diameter of the disk apertures is gradually decreased down the structure. For modern TW linacs, the phase advance per cavity is chosen to be $2\pi/3$ (i.e., three cavities per wavelength).

In a SW linac both ends of the structure are effectively shorted, so that electromagnetic waves are reflected back and forth resulting in a standing wave pattern (as in a single resonant cavity). Both TW and SW structures can be thought of as series of individual resonators that are electromagnetically coupled. A TW linac is capacitively coupled through the disk apertures, which must be large and thin to obtain adequate coupling. In contrast, the SW linac uses inductive coupling through peripheral slots from one cavity to its neighbors. The spatial concentration of the electric field at the electron bunch is therefore better in the SW structure, and the transit time characteristics are usually somewhat more favorable. As a result, the rf efficiency is significantly higher for a SW waveguide for the same accelerating gradient, although the SW cavity surfaces must be machined with greater precision.

The TW structure is somewhat less complex, and usually less expensive per unit length to fabricate. It does not require an isolator or circulator, since it is a matched device, but it does require both input and output couplers. The larger radius apertures permit a somewhat higher beam current. On the other hand, since the cavities are much more tightly coupled, the SW accelerator is much more stable in phase with respect to temperature variations, and has much less tuning sensitivity. For applications in which physical space and rf power efficiency are important, and beam stability is essential, the SW accelerator offers a number of advantages, and it has become the linac structure of choice for most food irradiation applications. In the remainder of this section we discuss SW structures in some detail. For more details on TW structures, there are several excellent references for the interested reader.^{6,7,11}

6.4.2. Inductively-Coupled Linac Structures

Most of the important features of an inductively coupled SW structure can be understood by extending the circuit model developed for the resonant cavity to the case of $N + 1$ coupled oscillators, with the currents in each oscillator corresponding to the fields in a cavity. To demonstrate the use of this model, consider the case of three coupled oscillators, shown in Figure 6-16.⁷ The center oscillator (cavity or cell) has a total inductance of $2L_o$ and capacitance C_o , and a resonant

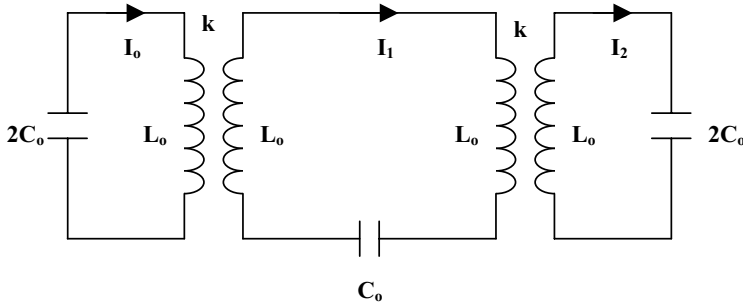


Figure 6-16. Modeling of three coupled cavities by three coupled resonator circuits.

frequency $\omega_o = (2L_o C_o)^{-1/2}$. The end oscillators (cavities) have half the inductance, but twice the capacitance.* The inductive slots that couple the cavities are modeled by a coupling constant $k = M/L_o$, with M being the mutual inductance.

Normalizing the current (field) I_n in the n^{th} oscillator according to $i_n = (2L_o)^{1/2} I_n$, and assuming a harmonic time dependence of the form $e^{j\Omega t}$, Kirchoff's equations for the three coupled circuits are

$$i_o(1 - \omega_o^2/\Omega^2) + i_1 k = 0 \quad (6.36)$$

$$i_1(1 - \omega_o^2/\Omega^2) + (i_o + i_2)(k/2) = 0 \quad (6.37)$$

$$i_2(1 - \omega_o^2/\Omega^2) + i_1 k = 0 \quad (6.38)$$

These three coupled equations can be written in matrix form as $AX_m = \Omega_m^{-2} X_m$, with the matrix operator A given as

$$A = \begin{bmatrix} 1/\omega_o^2 & k/\omega_o^2 & 0 \\ k/(2\omega_o^2) & 1/\omega_o^2 & k/(2\omega_o^2) \\ 0 & k/\omega_o^2 & 1/\omega_o^2 \end{bmatrix} \quad (6.39)$$

The eigenvectors X_m are given as

$$X_m = \begin{pmatrix} i_o \\ i_1 \\ i_2 \end{pmatrix} \quad (6.40)$$

and Ω_m^{-2} denote the corresponding eigenvalues. The three normal mode eigenfrequencies and their normalized eigenvectors are given by (in order of increasing frequency)

(1) $m = 0$; the zero mode, in which all oscillators have the same phase

$$\Omega_o = \omega_o(1 + k)^{-1/2}; \quad X_o = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad (6.41)$$

* In an accelerating structure the end oscillators would correspond to "half-cells" with conducting planes placed at the symmetry plane of a full accelerating cavity, as suggested in Figure 6-13.

- (2) $m = 1$; the $\pi/2$ mode, with a $\pi/2$ phase shift from oscillator to oscillator; note that the center oscillator is unexcited

$$\Omega_1 = \omega_0; \quad X_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \quad (6.42)$$

- (3) $m = 2$; the π mode, with a π phase shift from oscillator to oscillator

$$\Omega_2 = \omega_0(1 - k)^{-1/2}; \quad X_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \quad (6.43)$$

This same calculation can be extended in straightforward fashion to a system of $N + 1$ coupled oscillators, consisting of $N - 1$ identical internal oscillators with half-cells on each end. In this case, the relative magnitudes of the eigenvector elements of the m^{th} mode in the n^{th} oscillator are

$$X_{m,n} = \cos(\pi mn/N); \quad n = 0, 1, \dots, N \text{ (oscillator) and } m = 0, 1, \dots, N \text{ (mode)} \quad (6.44)$$

and the eigenfrequency of the m^{th} mode can be expressed by the dispersion relation

$$\Omega_m = \omega_0[1 + k \cos(\pi m/N)]^{-1/2} \quad (6.45)$$

The argument of the cosine is the phase advance per oscillator. The frequency of the zero mode is $\omega_0(1 + k)^{-1/2}$, while the π mode ($m = N$) has the highest frequency $\omega_0(1 - k)^{-1/2}$. The frequency of the $m = N/2$ mode, the $\pi/2$ mode, is always ω_0 . Note that the difference in frequency between the zero mode and the π mode depends only on the coupling constant k . As the number of cavities increases, the normal mode frequencies become more closely spaced. The frequency separation between next nearest modes is always largest for the $\pi/2$ mode. The dispersion relation and the relative field amplitudes are shown in Figures 6-17 and 6-18 for the case of $N = 6$.

The coupled oscillator model can also be used to examine the effects of resistive power flow losses in the cavities. When such losses are included in the analysis, they introduce a relative phase shift (the power-flow phase shift) between adjacent oscillators for all modes except the $\pi/2$ mode. This phase shift depends inversely on the product of the quality factor Q and the coupling constant k . In the $\pi/2$ mode, however, the power flow does not produce this phase shift; rather, it produces a first-order excitation of the normally unexcited resonators, and a second-order decrease in amplitude in the excited resonators that increases with distance from the driven cavity. This power flow droop explicitly depends on $(kQ)^{-2}$.

When accelerating structures are manufactured, machining tolerances inevitably lead to slight variations in the resonant frequencies of individual cavity resonators. The effects of such frequency errors can be examined using perturbation theory, with the results being expressed in terms of an unperturbed eigenvector for a particular mode, plus small corrections from the other unperturbed eigenvectors, that depend on the magnitude of the errors and the coupling constant. The results of such perturbation calculations indicate that frequency errors lead to first-order corrections in the magnitudes of the cavity fields for all modes except the $\pi/2$ mode. In contrast, frequency errors for this mode give rise to first-order field corrections in only the resonators that are normally unexcited. The field errors in the normally excited resonators are of second order.

In summary, the $\pi/2$ mode has several attractive features for particle acceleration. These include power flow insensitivity, insensitivity to frequency errors, and good mode separation. However, if all

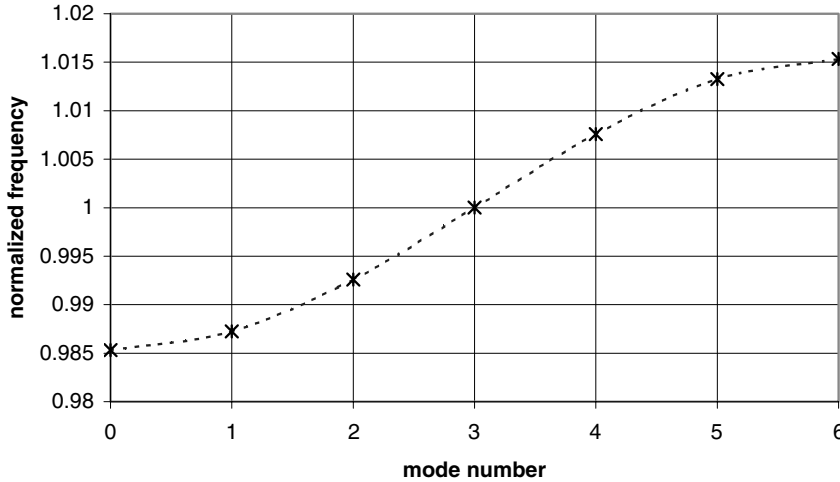


Figure 6-17. Dispersion relation for the case of $N + 1$ coupled oscillators with $N = 6$ and a coupling constant $k = 0.03$. The resonant frequency is normalized to f_0 .

the cavity resonators are identical, either the gradient or the shunt impedance must decrease because the accelerating fields exist in only half the available length of the structure (Figure 6-18).

A practical way of implementing the $\pi/2$ mode is to form a biperiodic chain of cavities in which the geometry of the excited cavities is optimized for high shunt impedance, while the unexcited (coupling) cavities are designed to occupy a small amount of axial space (but still have the same resonant frequency). The two most popular such accelerator structures are the on-axis coupled linac¹² and the side-coupled linac,¹³ shown schematically in Figure 6-19. For the on-axis structure the coupling cavities simply have a very short axial length; in the side-coupled structure, the coupling cavities are displaced from the axis. Both structures use magnetic coupling through slots cut in the cavity walls.

The coupled circuit model can also be used to analyze this biperiodic structure. Assuming N coupling cavities with resonant frequency ω_c interspersed in alternate fashion with $N + 1$ accelerating cavities with resonant frequency ω_a , the resulting dispersion relation can be written as⁷

$$k^2 \cos^2(m\pi/2N) = [1 - (\omega_a/\Omega_m)^2][1 - (\omega_c/\Omega_m)^2]; m = 0, 1, \dots, 2N \quad (6.46)$$

k is now the coupling constant between a coupling cell and an accelerating cell, and Ω_m is the frequency of the m^{th} normal mode. For the $\pi/2$ mode ($m = N$), the right-hand-side of Eq. (6.46) is zero, and there are two solutions: $\Omega_N = \omega_a$, corresponding to excited accelerating cells; and $\Omega_N = \omega_c$, corresponding to excited coupling cells. Since this latter mode would correspond to unexcited end cells, it is not allowed for physical reasons.

The normal mode spectrum of a coupled cavity structure is obtained by exciting the structure with a swept-frequency signal and detecting the field in an end cell. A typical result for $\omega_c < \omega_a$ is shown in Figure 6-20 for a biperiodic structure consisting of seven accelerating cavities and six coupling cavities. The corresponding dispersion relation is shown in Figure 6-21. The frequency ranges above and below the discontinuity are termed the upper and lower pass bands. The region between ω_a and ω_c is termed the stopband, within which there are no solutions.

This case of $\omega_a > \omega_c$ is usually to be avoided because it can lead to thermal instability. The stopband can be eliminated by designing (or tuning) the cavities such that $\omega_a = \omega_c$, a condition

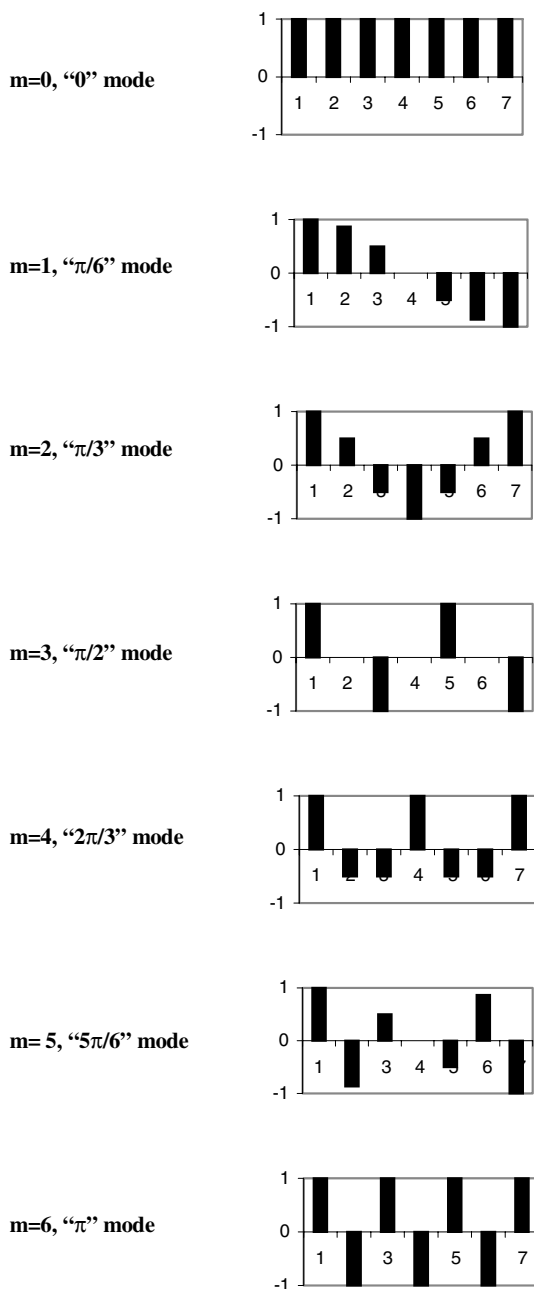


Figure 6-18. Relative amplitudes of the axial fields in a linac structure consisting of seven coupled cavities.

termed confluence. When this occurs, the slope of the dispersion curve increases to a maximum in the vicinity of the $\pi/2$ resonance, and the power-flow phase shift again vanishes. The dispersion relation for a biperiodic structure with 12 coupling cells and 13 accelerating cells in confluence is shown in Figure 6-22.¹⁴

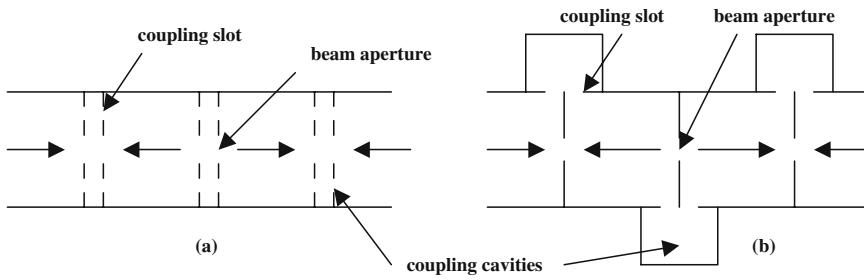


Figure 6-19. Schematic diagram of biperiodic SW accelerating structures: (a) axial-cavity coupling; (b) side-cavity coupling.

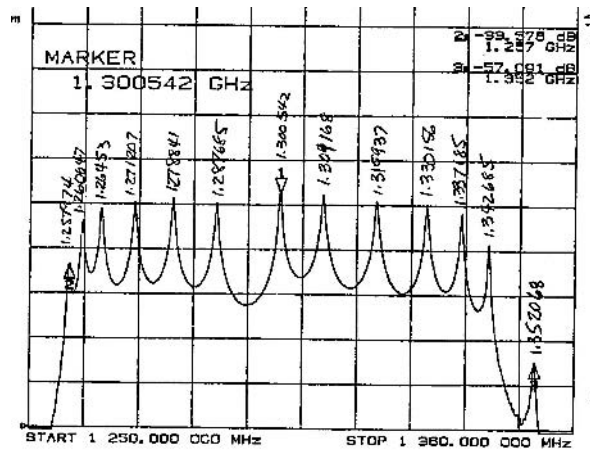


Figure 6-20. Normal mode spectrum for a 13-cell biperiodic structure with $\omega_c < \omega_a$.

Beta 1.3 GHz 13 cell 6.5 accelerating cells

w1= 1.30193784 ,w2= 1.29652448 ,gap=-0.00404926 ,chi2= 1339.0

k= 6.75661E-02 ,kk= -2.09879E-03

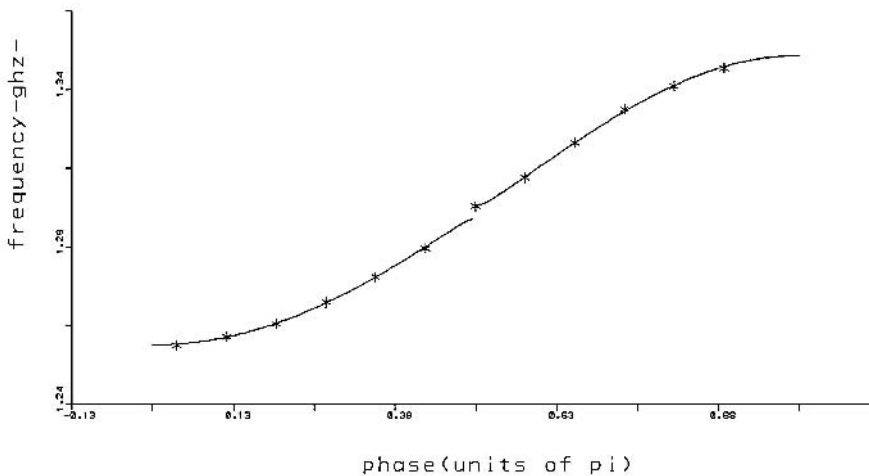


Figure 6-21. Dispersion relation for the 13-cell structure of Figure 6.20 with $\omega_c < \omega_a$. Note the 4-MHz stopband; the coupling constant is approximately 6.7%. The coupling constant between adjacent accelerating cells is only 0.2%.

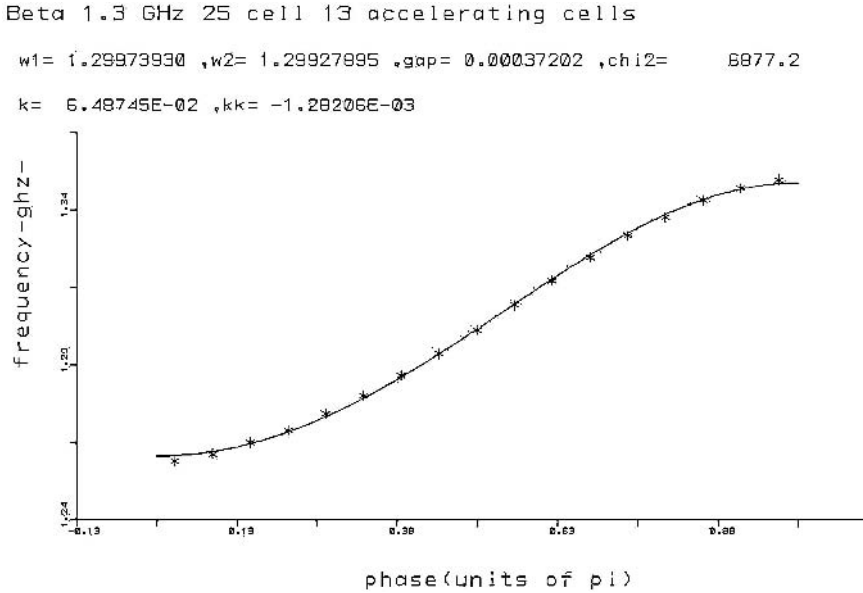


Figure 6-22. Dispersion relation for a 25-cell biperiodic structure that has been brought into confluence by tuning ($\omega_c = \omega_a$). There is no stopband, and the slope of the dispersion relation is largest in the vicinity of the $\pi/2$ resonance.

For relativistic SW structures the wave phase velocity (determined by the length of an accelerating/coupling cavity pair) is essentially equal to the speed of light. However, the group velocity is determined from the slope of the dispersion curve at the operating frequency. Since the group velocity is the velocity with which microwave energy propagates down the structure, operating in the $\pi/2$ mode with a large coupling constant leads to rapid energy propagation. Consequently, when power is coupled into a SW linac, the fields in all cavities tend to increase nearly simultaneously, rising to the saturated value in a fill time approximately determined by

$$\tau_{\text{fill}} = 2Q/\Omega_{\pi/2} \quad (6.47)$$

Practical embodiments of the two most common types of coupled cavity linac structures are shown in Figure 6-23.^{6,14} The length of a complete acceleration block, consisting of a single accelerating cell and coupling cell, is equal to the distance that an electron travels in one-half the microwave period. Such blocks are brazed together to form accelerating structures of the desired length. The side-coupled linac typically has higher shunt impedance, but the axially-coupled structure is usually less expensive to fabricate. The axially-coupled linac can have a larger coupling coefficient and is therefore more stable and more easily tuned.

6.5. COUPLING TO RESONANT ACCELERATOR STRUCTURES⁷

The three methods for introducing radio-frequency power into a cavity are schematically shown in Figure 6-24. These include (1) a magnetic coupling loop, (2) an electric coupling antenna at the end of a coaxial transmission line, and (3) an iris opening in the cavity wall to which a waveguide is attached. For rf accelerators, the magnetic coupling loop is most commonly used for low-frequencies, while the waveguide iris is most common at microwave frequencies. While not usually used for

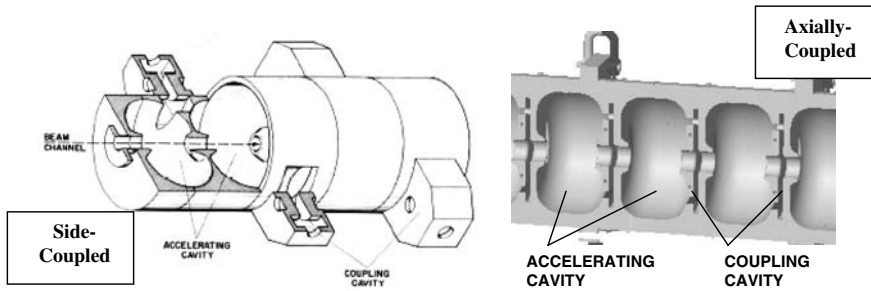


Figure 6-23. Perspective drawings of side-coupled and axially-coupled SW accelerator structures. (The drawing of the side-coupled structure is reprinted courtesy of the Los Alamos National Laboratory.)

acceleration (because the probe would normally occupy the region of the beam), electric field coupling is nevertheless quite useful for determining resonant frequencies and quality factors under cold-test (low-power, no beam) conditions.

6.5.1. Coupling with Minimal Beam Loading

The external circuit that drives the cavity and the cavity itself can be described by extending the parallel RLC circuit model to include an external driving circuit consisting of a matched current generator, a transmission line with characteristic impedance Z_0 , and a coupling mechanism represented by a transformer with a turns ratio of 1:n. This analysis is especially useful for the loop coupling method. For a resonant structure coupled to a waveguide, it is more conventional to define a general parameter β_0 that is a measure of the waveguide-to-cavity coupling strength in the absence of any beam loading. This parameter is given by

$$\beta_0 = Q_0/Q_{ex} = P_{ex}/P_c \quad (6.48)$$

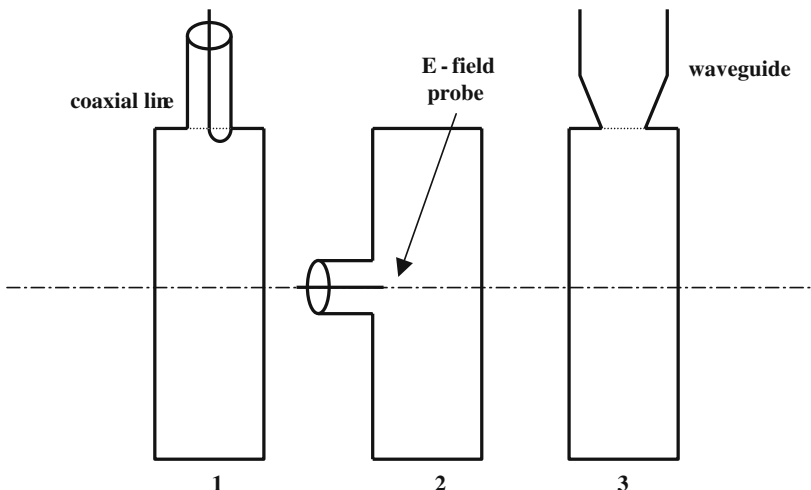


Figure 6-24. Techniques for coupling rf power to cavities: (1) loop coupling (magnetic) from a coaxial cable; (2) antenna (electric) coupling from a coaxial cable; and (3) waveguide-iris coupling.

As before, P_c is the power loss in the cavity; the external power P_{ex} is the power that would be absorbed in the driving circuit if the microwave source were turned off. The quality factor associated with P_c is termed the external Q , Q_{ex} . With the generator turned off, the external circuit loads the cavity, and the average power dissipated is $P = P_{ex} + P_c$. Consequently, the loaded Q is defined as

$$Q_L = \omega_o U / (P_{ex} + P_c) = Q_o Q_{ex} / (Q_o + Q_{ex}) = Q_o / (1 + \beta_o) \quad (6.49)$$

For a cavity driven at resonance, the load impedance of the driving circuit is $Z_L = \beta_o Z_o$, and the resulting reflection coefficient of the input coupler is given as

$$\Gamma = (Z_L - Z_o) / (Z_L + Z_o) = (\beta_o - 1) / (\beta_o + 1) \quad (6.50)$$

If P_t is the total incident power from the generator, then the power reflected from the input coupler backward toward the generator is

$$P_r = \Gamma^2 P_t = [(\beta_o - 1) / (\beta_o + 1)]^2 P_t \quad (6.51)$$

and the power delivered to the cavity through the input coupler is

$$P_c = P_t (1 - \Gamma^2) = 4\beta_o P_t / (1 + \beta_o)^2 \quad (6.52)$$

When $\beta_o = 1$, the cavity is critically coupled to the generator and $P_c = P_t$.

The voltage associated with the steady-state power loss in the cavity can be considered to be produced by the sum of a forward-going voltage wave V_+ and a backward going wave V_- , with $V_- = \Gamma V_+$. Assuming that V_+ is constant at $t = 0$, then we can introduce the known time dependence of the cavity voltage from Eq. (6.33), and the time-dependent reflection coefficient can be written as

$$\Gamma(t) = V_-(t) / V_+ = [2\beta_o / (1 + \beta_o)] (1 - e^{-t/\tau}) - 1 \quad (6.53)$$

with τ now defined in terms of the loaded Q as $\tau = 2Q_L / \omega_o$. The variation of Γ^2 with time is graphed in Figure 6-25. If $\beta_o = 1$, then Γ^2 monotonically increases to 0, at which time there is no backward-going wave. To reach the maximum steady-state cavity voltage, the input voltage must be longer than typically 2τ . For the undercoupled case, $\beta_o < 1$, the reflected power again decreases monotonically, but to a non-zero value in the steady state. When $\beta_o > 1$, the system is overcoupled; the reflected power first decreases to zero, then increases to its steady-state value. Also note that at $t = 0$, $\Gamma = -1$ always, implying that all of the incident power is reflected. To prevent damage to the generator, even for a well-matched system, it is usually desirable to insert a circulator or other isolator device between the cavity and the generator so that any reflected power is diverted into a resistive load.

6.5.2. Optimum Coupling with Beam Loading

It is usually desirable to design the input coupler geometry to minimize reflected power in the steady-state. However, when significant current is accelerated, the passage of the beam through the structure can induce fields comparable to those produced by the microwave source, and the net accelerating field is the superposition of the source field and the beam-induced field. This beam loading effect is especially important in ion accelerators for which longitudinal focusing is necessary, implying that the synchronous phase angle (between the beam bunches and the peak of the accelerating field)

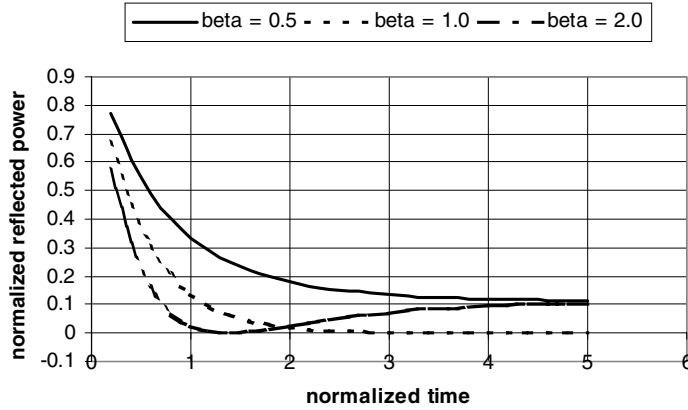


Figure 6-25. Reflected power versus time for different values of the coupling factor β_0 .

is not zero. In this case it is also necessary to detune the frequency of the source from that of the accelerating structure to achieve minimum reflected power.

For relativistic electron accelerators, longitudinal focusing is not required, and the electron bunches can be accelerated at the crest of the accelerating field for maximum energy gain. In this case the structure is driven at its resonant frequency, the cavity impedance seen by the source is purely real, and the source power is increased to offset the voltage reduction induced by the beam. The input coupler geometry must therefore be designed to account for the increased power flow.

To account for beam loading we can modify Eq. (6.48) as

$$\beta = P_{ex}/(P_c + P_b) \quad (6.54)$$

in which P_b represents the power that goes into the beam. Eliminating P_{ex} gives

$$\beta = \beta_0(1 + P_b/P_c)^{-1} \quad (6.55)$$

The total power required from the source P_t is defined from energy conservation as

$$P_t = P_r + P_b + P_c \quad (6.56)$$

with the reflected power $P_r = \Gamma^2 P_t$. To minimize the required generator power at full beam loading we should choose $\beta = 1$, or from Eq. (6.55)

$$\beta_0 = 1 + P_b/P_c \quad (6.57)$$

The efficiency with which an accelerating structure delivers microwave power to the beam is simply P_b/P_t . As a numerical example, we will assume that the structure efficiency is 75% when the input coupler is matched, implying that $P_b = 3P_c$. Minimum reflected power will then occur when the input coupler has an unloaded coupling factor of $\beta_0 = 4$.

We now assume that the input coupler has been brazed to the accelerating structure, but that the beam current is not equal to the design value. Taking into account beam loading, the reflection

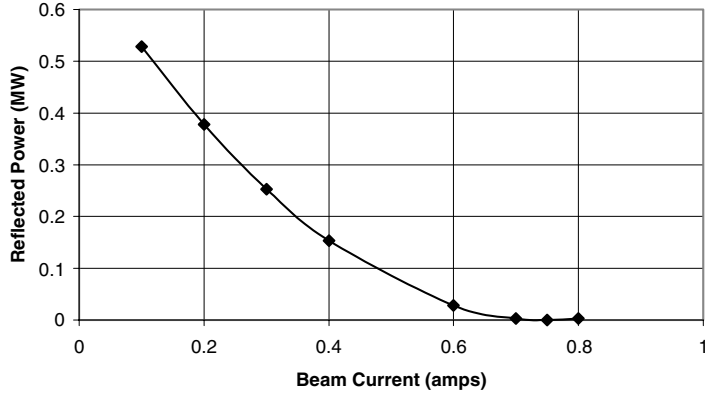


Figure 6-26. Reflected power vs beam power for a 5-MeV linac assuming a coupling parameter $\beta_0 = 4$, and fixed cavity losses of 1.25 MW.

coefficient Γ is now given in terms of the beam-loaded β as

$$\Gamma = [(1 - \beta)/(1 + \beta)] \quad (6.58)$$

and the reflected power is given by

$$P_r = [\Gamma^2/(1 - \Gamma^2)](P_b + P_c) \quad (6.59)$$

The microwave power reflected from the coupler as a function of the beam current is shown in Figure 6-26 for the case corresponding to $\beta_0 = 4$, $P_c = 1.25$ MW, and a fixed kinetic energy of 5 MeV, as an illustrative example. A perfect match corresponds to the case of 0.75 amps and a beam power of 3.75 MW. The match is quite good over a beam current variation of more than 100 mamps, however.

6.5.3. Measurement of the Coupling Parameter

The coupling parameter β_0 can be measured by detecting the reflected signal when power is launched through the waveguide toward the coupler. The results are usually displayed on a “Smith chart,” which consists of a set of constant impedance (constant β) circles that pass through the point $(-1, 0)$.¹⁵ The radii of these circles are given by $R_Z = 1/(1 + Z_L/Z_0) = 1/(1 + \beta)$. The outer circle of the Smith chart is defined by $R_Z = 1$, corresponding to $\beta = 0$, which defines a short circuit. For a matched load, $\beta = 1$ and $R_Z = 0.5$.

If the frequency of the input power is swept through a resonance of the structure under test, then a “resonance circle” will appear inside the short circuit impedance circle of the Smith chart. The radius r_c of the resonance circle determines the coupling parameter for the measurement according to

$$\beta_m = r_c/(1 - r_c) \quad (6.60)$$

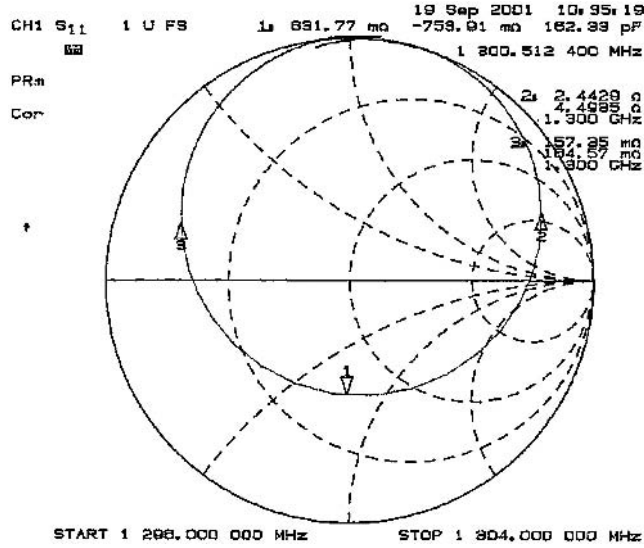


Figure 6-27. Smith chart measurement of the coupling beta for a waveguide-to-cavity transition. The radius of the resonance circle is approximately 0.74, giving $\beta_m = 2.8$.

The unloaded coupling parameter is then inferred from the measured value by correcting for the loaded quality factor, according to

$$\beta_o = [\beta_m / (1 + \beta_m)](Q_o / Q_L) \quad (6.61)$$

An example of this procedure is shown in Figure 6-27. For a loaded Q_L of 3645, the measured coupling parameter was $\beta_m = 2.8$. With an unloaded Q_o estimate of 17000 (from measurements and Superfish calculations), β_o was inferred from Eq. (6.61) to be 3.44.¹⁴

6.5.4. Load Line of a SW Structure

Representing the beam power as $P_b = VI_b$, with I_b the beam current, and $P_c = V^2/R$, with R being the total shunt impedance of the structure, Eq. (6.55) can be written as

$$\beta = \beta_o(1 + RI_b/V)^{-1} \quad (6.62)$$

Solving Eq. (6.58) for Γ^2 and substituting the result into Eq. (6.56) gives an expression for the load line of the structure⁶

$$V = [2(\beta_o RP_t)^{1/2} - I_b R] / (1 + \beta_o) \quad (6.63)$$

The kinetic energy of the accelerated beam therefore increases with the square root of the power from the source, and decreases linearly with the accelerated beam current. As a numerical example, we assume $\beta_o = 4$, a shunt impedance per unit length of 75 MΩ/m, and a structure length of one meter. Then $R = 75 \text{ M}\Omega$, and Eq. (6.63) becomes

$$V = 6.93 \times 10^3 P_t^{1/2} - 1.5 \times 10^7 I_b$$

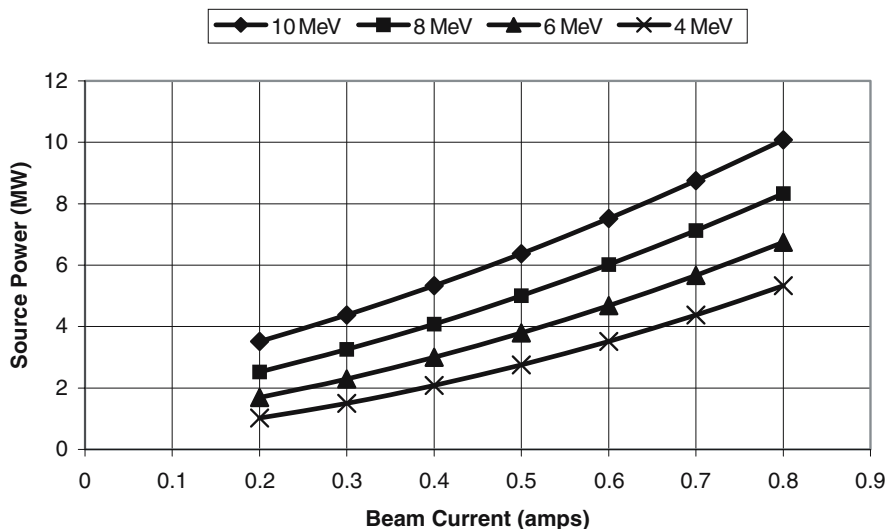


Figure 6-28. Source power required to produce a particular beam kinetic energy as a function of accelerated beam current, for $\beta_0 = 4$ and $R = 75 \text{ M}\Omega$.

Lines of constant kinetic energy for this load line are shown in Figure 6-28 versus source power and beam current. For a 5-MW source, and a macropulse current of 0.5 amps, the beam kinetic energy would be approximately 8 MeV for this linac.

6.6. SINGLE-CAVITY RF ACCELERATORS

Having discussed several essential concepts used in microwave accelerators, we will next illustrate their use with numerical simulation examples. In this section we will examine the simplest microwave accelerator consisting of an electron gun and a single accelerating cavity through which the beam makes a single pass. The results will therefore be somewhat representative of an ILU accelerator (see Section 5.2.3).¹⁶ In the following section we will examine the case of a multiple-cavity microwave linac.

We assume the cavity is driven in the TM_{010} mode by a high-power triode at a frequency of 175 MHz, and adopt nominal design goals of 50 kW of average beam power at a maximum beam kinetic energy of 5 MeV, implying an average beam current of 0.01 amps. With an assumed peak power of 1 MW from the triode, and assuming a cavity efficiency of 50% ($P_b = P_c$), the peak macropulse beam current should be 0.1 amp, which implies a 10% duty cycle, e.g., 1 msec at 100 Hz.

The accelerating cavity will be a full cavity, with halves of the general shape shown in Figure 6-13; the electron gun is placed in one of the nose cone apertures. From Eq. (6.22), the outer radius of the cavity will be approximately 65 cm. The accelerating gap design should have the highest transit time factor consistent with the maximum field stress from the Kilpatrick criterion, Eq. (6.35). At $f = 175 \text{ MHz}$, $E_k = 14 \text{ MV/m}$. Because of the millisecond pulse duration, we assume a bravery factor of only 1.25, implying that the peak field stress should not exceed 17.5 MV/m on the nose cones. A cavity shape satisfying these criteria is shown in Figure 6-29. The distance between electrodes is 38 cm, and the outer cavity radius is 59 cm. The ratio of the peak field on the nose cone to the average field over the 70-cm length of the cavity is 2.98.

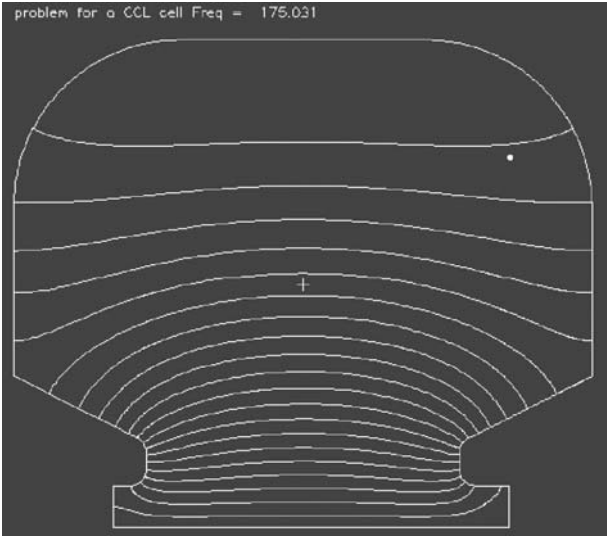


Figure 6-29. Design of the single acceleration cavity with a resonant frequency of 175 MHz. The outer cavity radius is 59 cm and the total cavity length is 70 cm. The distance between nose cones is 38 cm, and the nose cone aperture radius is 5 cm. The ratio of the peak field on the nose cone to the average field over the 70-cm length is 2.98.

With a constant voltage applied to the gun, we might expect the sinusoidal accelerating fields of the cavity to capture approximately one-half of the emitted current, implying that the total gun current should be about 0.2 amps. It is instructive to examine the capture and acceleration processes in some detail. The phase space trajectories of test electrons crossing the acceleration gap are plotted in Figure 6-30, assuming a sinusoidally varying electric field with amplitude of about 15 MV/m. The test electrons were emitted uniformly in time over one complete rf cycle ($-\pi, \pi$). A gun voltage of 15 kV was assumed.

Only those electrons emitted with relative phase angles in the range of $(-0.25, 2.34)$ cross the gap, implying a capture efficiency of a little more than 40%. Those electrons emitted with relative phase angles around zero tend to be axially focused, forming a sharp leading edge of the accelerated

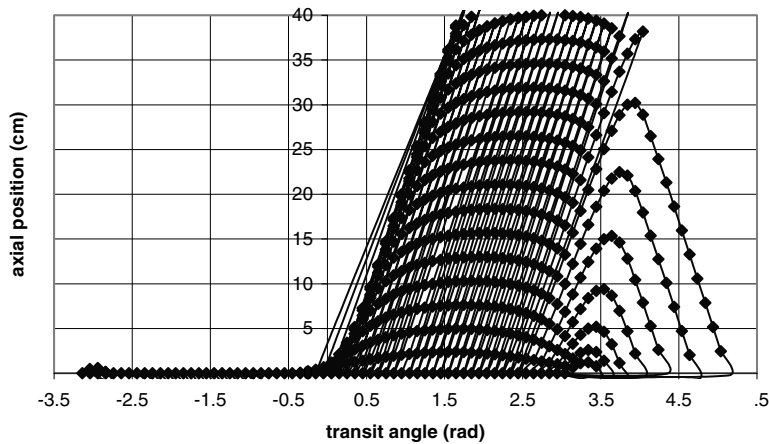


Figure 6-30. Phase space trajectories of trial electrons crossing the accelerating gap of the cavity shown in Figure 6-29.

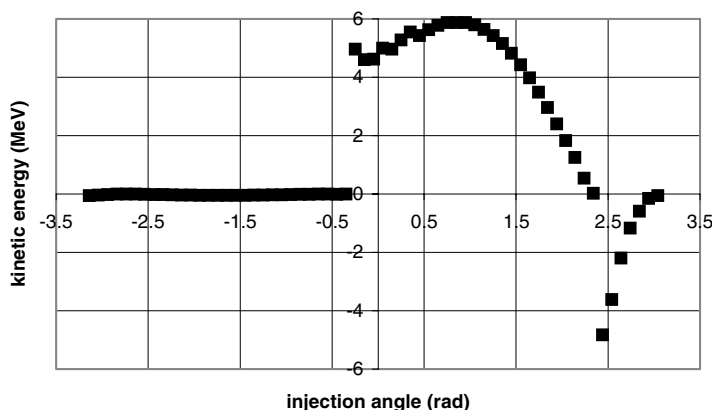


Figure 6-31. Kinetic energies of transmitted and reflected electrons as a function of their injection phase angle relative to the sinusoidal accelerating field.

bunch. The kinetic energies of the transmitted and reflected electrons are plotted as a function of relative phase angle at entry to the gap in Figure 6-31. (Negative energies denote electrons reflected backward.) There is a large energy spread in the transmitted electrons (0–6 MeV), and some of the reflected electrons will impact the cathode with considerable energy.

To avoid these problems of excessive cathode back-bombardment and a poor kinetic energy spectrum, single cavity rf accelerators almost always use some form of gridded-gun arrangement that permits control of the gun conduction angle. That is, electrons are only emitted when they can be efficiently captured and accelerated by the cavity fields.*

6.7. MULTIPLE-CAVITY RF ACCELERATORS

To illustrate the interplay of various multiple-cavity linac parameters, we will next examine an S-band linac at 2.856 GHz having the nominal performance parameters summarized in Table 6.3. The peak kinetic energy is 10 MeV, with an average beam power of 15 kW at a duty cycle of nominally 0.5% (e.g., 20 μ sec at 250 Hz). Assuming an average kinetic energy of 9.5 MeV, the average beam current should be approximately 1.6 mamps, and the beam current during the macropulse should be in the range of 0.3–0.35 amperes, corresponding to a useful peak beam power of approximately 3 MW. Assuming a capture efficiency of 0.4–0.5 from a standard diode gun, the gun current should be in the range of 0.6–0.8 amperes. Beam power losses due to reflection and nose cone interception are estimated at 10% of the useful beam power, or 0.3 MW. If the structure efficiency is 70%, then the copper power losses will be approximately 1.4 MW, and the input coupler should therefore have an unloaded coupling parameter of $\beta_0 = 3.35$. A klystron with a peak power rating of 5 MW should be sufficient. With a gradient of 10 MeV/m, the structure should be approximately one meter in length. At 2.856 GHz, the accelerating block length for a relativistic electron is approximately 5.25 cm, implying that there should be about 19 accelerating cavities.

To simulate the performance of this hypothetical linac we will use a fixed-frame version of *Parmela*.¹⁷ Required inputs include the gun design (from *EGUN*), the suitably scaled fields of the accelerating cavities (from *SUPERFISH*), and (if necessary) any magnetic fields to assist beam

* In fact, the ILU accelerators use a unique gridded gun configuration in which some of the cavity fields are deliberately allowed to leak into the space between cathode and grid.¹⁶

Table 6.3. Multiple-Cavity Linac Parameter Goals

Parameter	Value
Maximum kinetic energy	10 MeV
Average energy	9.5 MeV
Macropulse current	0.32 amps
Efficiency (microwave to e-beam)	70%
Duty cycle (pulse duration x rep rate)	0.005
Average e-beam power	15 kW
Accelerating gradient	10 MeV/m

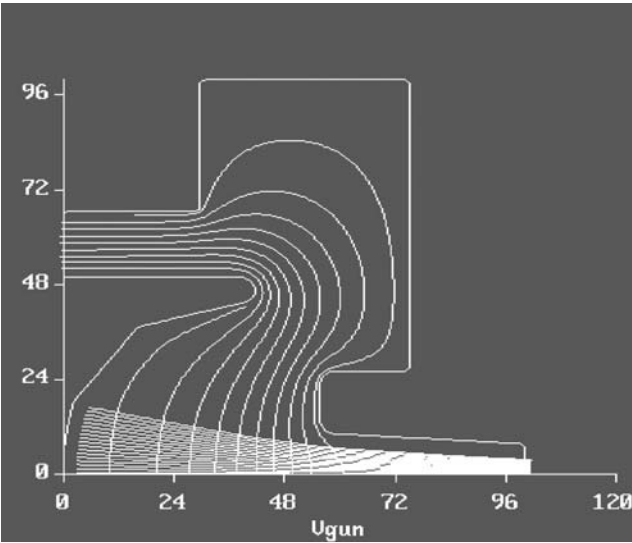


Figure 6-32. Electron flow trajectories for the electron gun. With 25 kV applied voltage, the gun generates 0.8 amps.

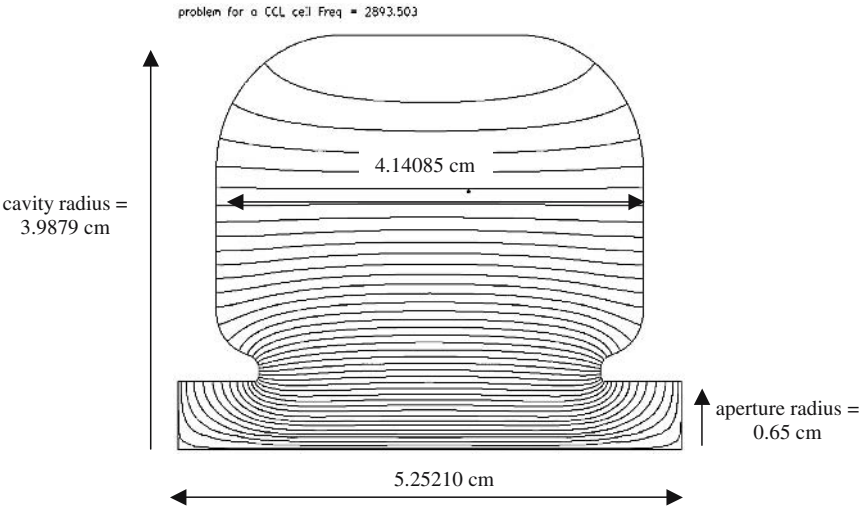


Figure 6-33. Relativistic accelerating cavity design for the 2.856-GHz linac.

Table 6.4. SUPERFISH Relativistic Cavity Output Data

Parameter	Beta = 1	Half-cell	Int. Beta = 0.85
Transit Time Factor (T)	0.77	0.82	0.81
Quality Factor (Q)	18331	9827	15207
Shunt Impedance per Unit Length	142.5 MΩ/m	40.9 MΩ/m	110.3 MΩ/m
ZT ²	84.0 MΩ/m	27.8 MΩ/m	72.2 MΩ/m
Peak to Average Field Ratio	3.45	5.80	3.83

transport (from POISSON). Assuming a capture efficiency of 40%, the gun must supply 0.8 amps. An EGUN calculation for a gun design having a perveance of 2×10^{-7} is shown in Figure 6-32. The required voltage is 25 kV.

At a frequency of 2.856 GHz, the center-to-center length of an accelerating/coupling cavity block for a relativistic electron is 5.2521 cm. Assuming that such a cavity provides an average energy gain of about 0.5 MeV, the average field is about 10 MV/m. Since the Kilpatrick criterion is $E_k = 46$ MV/m at 2.856 GHz, the peak-to-average field stress could exceed six with even a low bravery factor. A cavity design satisfying these modest criteria is shown in Figure 6-33. The interior cavity width of 4.14085 cm permits use of axial coupling cavities, if desired.

The half-cell used to capture the beam from the gun is shown in Figure 6-34. Since the energy gain from this half-cell will be no greater than about 0.2 MeV, a cavity design similar to that of Figure 6-33, but for an intermediate electron velocity of 0.85c, is used as the second cavity. A summary of the SUPERFISH output parameters for these three cavity designs is presented in Table 6.4.

The transit time factors apply to the entire cavity length; e.g., the transit time factor for the gap region of the relativistic cavity between the nose cones is about 0.85. Assuming 5% coupling, the shunt impedances are reduced by 15% to compute the copper power loss.

The simulation used a total of 6000 macroparticles uniformly injected over four complete rf cycles. Summary results of the simulation (based on the middle two rf cycles) are given in Table 6.5; they are very nearly equal to the nominal performance goals. Various frames from the simulation are

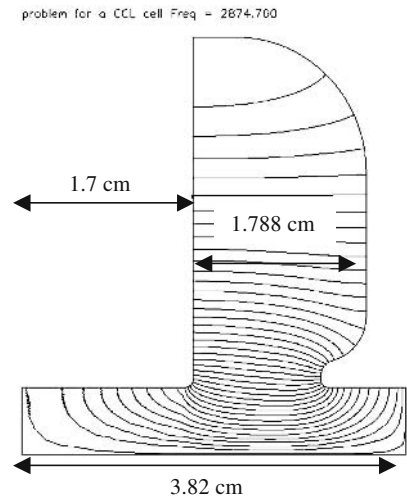


Figure 6-34. Half-cell design for the S-band linac.

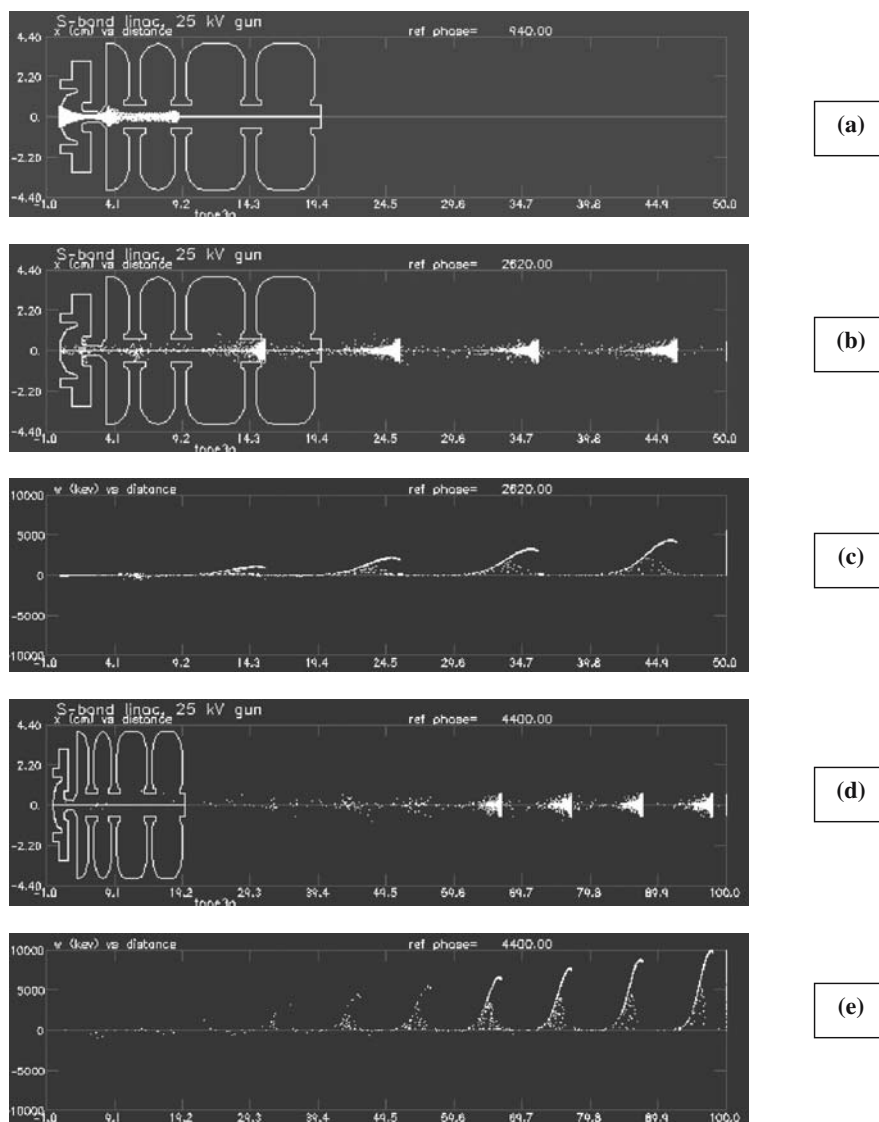


Figure 6-35. Frames from the numerical simulation of the S-band linac. (a) first half (50 cm) of the linac at relative phase 940 showing electron reflection at the half-cell; (b) first half of the linac at relative phase 2620 showing four well-formed bunches; (c) energy vs axial position for the first half of the linac; (d) full linac at relative phase 4400; (e) energy vs axial position for the full linac. (The final 15 accelerating cells are not shown.)

shown in Figures 6-35 and 6-36. Note the focusing of the beam from the gun into the first half-cell, and the increased radius of the charge at the time of reflection from the half-cell. Somewhat later there are four well defined bunches; each bunch has a sharp leading edge (also suggested by Figure 6-30 for the single cavity), with the highest kinetic energy occurring just behind the leading edge. The axial profile of a bunch remains constant after the first two accelerating cells as it is systematically accelerated down the linac; the peak kinetic energy at the exit is nearly 10 MeV.

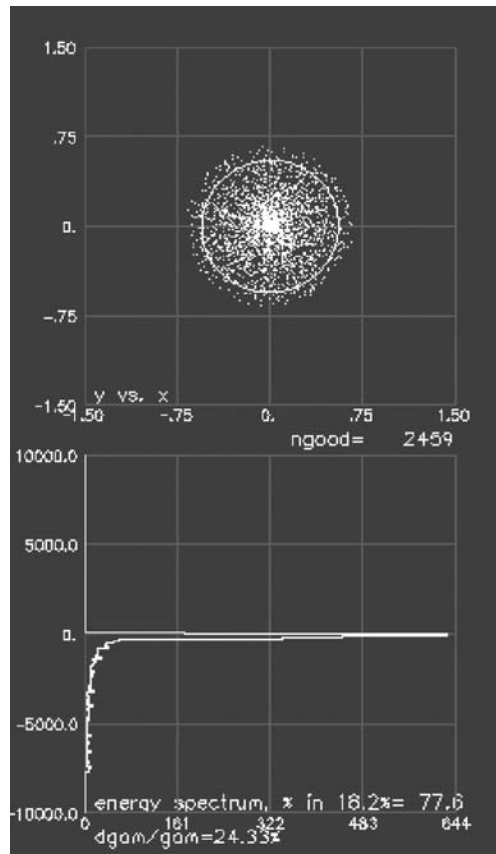


Figure 6-36. Beam x-y coordinate plot and kinetic energy spectrum at the end of the linac.

The x-y coordinates of the particles and the beam kinetic energy spectrum (relative to a synchronous particle in the bunch) at the linac exit are shown in Figure 6.36. The beam kinetic energy is sharply peaked at 9.85 MeV. The beam spot is not well defined and is essentially a truncated Gaussian. In fact, a tight beam spot with very high current density on axis is not desirable for foil heating and dose uniformity reasons. Similarly, very low beam emittance is not particularly important for food irradiation applications.

Table 6.5. Summary Simulation Results of the Hypothetical S-Band Linac

Parameter	Value
Maximum kinetic energy	9.91 MeV
Average kinetic energy	9.60 MeV
Beam current	0.32 amps
Beam power	3.1 MW
Total beam loading	3.4 MW
Copper power	1.5 MW

6.8. SUMMARY

In this chapter we have examined the essential concepts associated with the generation and acceleration of electron beams in microwave (radio-frequency) accelerators. The frequency range of interest is typically 100 MHz to 10 GHz, because electromagnetic waves with the corresponding wavelengths of 3 meters to 3 centimeters can be readily confined in metallic cavities of physically convenient size.

The electron beam in a microwave accelerator is almost always generated in a Pierce-type electron gun, using a thermionic cathode operated in a space-charge-limited mode. However, there are several particular types of accelerating structures, including both single and multiple-cavity designs (capacitively-coupled traveling wave and inductively-coupled standing wave linacs).

The individual cavities in these structures are usually designed to minimize resistive losses (as expressed by the shunt impedance) subject to electrical breakdown constraints. Shunt impedances in the range of 50–100 M Ω /m are now commonly achieved in standing-wave linacs, and this type of structure has become the standard microwave accelerator approach for food irradiation applications because of its compact size, efficiency, and stability.

The analysis of electromagnetic accelerating structures is facilitated by the use of resonant electrical circuit models. In particular, such models clearly indicate the advantageous properties of the $\pi/2$ mode for standing wave linacs. A practical method for operating in the $\pi/2$ mode is to construct a biperiodic structure consisting of accelerating cells designed for high shunt impedance, with alternating coupling cells that occupy minimum axial length.

Because the accelerating fields in microwave accelerators are sinusoidal, the kinetic energy gain of an electron depends on its phase angle with respect to the oscillating field and its transit time through the accelerating gap of the cavity. The important features of electron capture, bunch formation, and kinetic energy spread can be readily demonstrated and analyzed using numerical particle simulation techniques.

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